

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.11	Motor Nominal Speed	24	19200	RPM	Varies	112	Find this value n_n on the nameplate of the motor.
1.12	Motor Nominal Current	IH * 0.1	IH * 2	A	Varies	113	Find this value I_n on the nameplate of the motor.
1.13	Motor Cos Phi (Power Factor)	0.30	1.00		Varies	120	Find this value on the nameplate of the motor.
1.14	Energy Optimization	0	1		0	666	0 = Disabled 1 = Enabled
1.15	Identification	0	2		0	631	0 = No action 1 = At standstill 2 = With rotation
1.16	Start Function	0	1		0	505	0 = Ramping 1 = Flying Start
1.17	Stop Function	0	1		0	506	0 = Coasting 1 = Ramping
1.18	Automatic Reset	0	1		0	731	0 = Disabled 1 = Enabled
1.19	Response to External Fault	0	3		2	701	0 = No action 1 = Alarm 2 = Fault (Stop according to stop mode) 3 = Fault (Stop by coasting)
1.20	Response to AI Low Fault	0	5		0	700	0 = No action 1 = Alarm 2 = Alarm+preset fault frequency (P3.9.1.13) 3 = Alarm + previous frequency 4 = Fault (Stop according to stop mode) 5 = Fault (Stop by coasting)
1.21	Remote Control Place	0	1		0	172	0 = I/O control 1 = Fieldbus control
1.22	I/O Control Reference A Selection	1	20		6	117	1 = Preset Frequency 0 2 = Keypad reference 3 = Fieldbus 4 = AI1 5 = AI2 6 = AI1+AI2 7 = PID reference 8 = Motor potentiometer 11 = Block Out.1 12 = Block Out.2 13 = Block Out.3 14 = Block Out.4 15 = Block Out.5

Index	Parameter	Min	Max	Unit	Default	ID	Description
							16 = Block Out.6 17 = Block Out.7 18 = Block Out.8 19 = Block Out.9 20 = Block Out.10
1.23	Keypad Control Reference Selection	1	20		1	121	See P1.22.
1.24	Fieldbus Control Reference Selection	1	20		2	122	See P1.22.
1.25	AI1 Signal Range	0	1		0	379	0= 0...10 V/0...20 mA 1= 2...10 V/4...20 mA
1.26	AI2 Signal Range	0	1		1	390	0= 0...10 V/0...20 mA 1= 2...10 V/4...20 mA
1.27	RO1 Function	0	74		2	11001	See P3.5.3.2.1.
1.28	RO2 Function	0	74		3	11004	See P3.5.3.2.1.
1.29	RO3 Function	0	74		1	11007	See P3.5.3.2.1.
1.30	AO1 Function	0	31		2	10050	See P3.5.4.1.1.

Table 8: M1.34 Multi-pump (Single Drive)

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.34.1	PID Gain	0.00	100.00	%	100.00	118	
1.34.2	PID Integration Time	0.00	600.00	s	1.00	119	
1.34.3	PID Derivation Time	0.00	100.00	s	0.00	132	
1.34.4	Process Unit Selection	1	44		1	1036	See P3.13.1.4
1.34.5	Process Unit Min	Varies	Varies		Varies	1033	
1.34.6	Process Unit Max	Varies	Varies		Varies	1034	
1.34.7	Feedback 1 Source Selection	0	30		2	334	See P3.13.3.3
1.34.8	Setpoint 1 Source Selection	0	32		1	332	See P3.13.2.6
1.34.9	Keypad Setpoint 1	Varies	Varies	Varies	0	167	
1.34.10	SP1 Sleep Frequency Limit	0.0	320.0	Hz	0.0	1016	
1.34.11	SP1 Sleep Delay	0	3000	s	0	1017	0 = Not used
1.34.12	SP1 Wake Up Level	Varies	Varies	Varies	Varies	1018	0 = Not used
1.34.13	Multi-pump Mode	0	2		0	1785	0= Single drive 1= Multifollower 2= Multimaster
1.34.14	Number of Pumps	1	8		1	1001	
1.34.15	Pump Interlocking	0	1		1	1032	0 = Disabled

Index	Parameter	Min	Max	Unit	Default	ID	Description
							1 = Enabled
1.34.16	Autochange	0	2		1	1027	0 = Disabled 1 = Enabled (interval) 2 = Enabled (weekdays)
1.34.17	Autochanged Pump	0	1		1	1028	0 = Auxiliary Pump 1 = All Pumps
1.34.18	Autochange Interval	0.0	3000.0	h	48.0	1029	
1.34.19	Autochange Days	0	127			15904	B0 = Sunday B1 = Monday B2 = Tuesday B3 = Wednesday B4 = Thursday B5 = Friday B6 = Saturday
1.34.20	Autochange Time of Day	00:00:00	23:59:59	Time		15905	Range: 00:00:00-23:59:59
1.34.21	Autochange: Frequency Limit	0.00	P3.3.1.2	Hz	25:00	1031	
1.34.22	Autochange: Pump Limit	1	4			1030	
1.34.23	Bandwidth	0	100	%	10	1097	
1.34.24	Bandwidth Delay	0	3600	s	10	1098	
1.34.25	Pump 1 Interlock				DigIN Slot0.1	426	OPEN = Not active CLOSED = Active
1.34.26	Pump 2 Interlock				DigIN Slot0.1	427	See 1.34.25
1.34.27	Pump 3 Interlock				DigIN Slot0.1	428	See 1.34.25
1.34.28	Pump 4 Interlock				DigIN Slot0.1	429	See 1.34.25
1.34.29	Pump 5 Interlock				DigIN Slot0.1	430	See 1.34.25
1.34.30	Pump 6 Interlock				DigIN Slot0.1	486	See 1.34.25
1.34.31	Pump 7 Interlock				DigIN Slot0.1	487	See 1.34.25
1.34.32	Pump 8 Interlock				DigIN Slot0.1	488	See 1.34.25

3.5 Multi-pump (Multidrive) Application

Use the Multi-pump (multidrive) application in a system that has the maximum of 8 parallel speed motors with different speeds, for example, pumps, fans or compressors. By default, the Multi-pump (Multidrive) application is configured for 3 parallel motors.

The checklist for the commissioning a Multi-pump (multidrive) system is in [11.17.1.1 Commissioning Multi-pump \(Multidrive\)](#).

Each motor has a drive controls that applicable motor. The drives of the system communicate with each other by Modbus RTU communication.

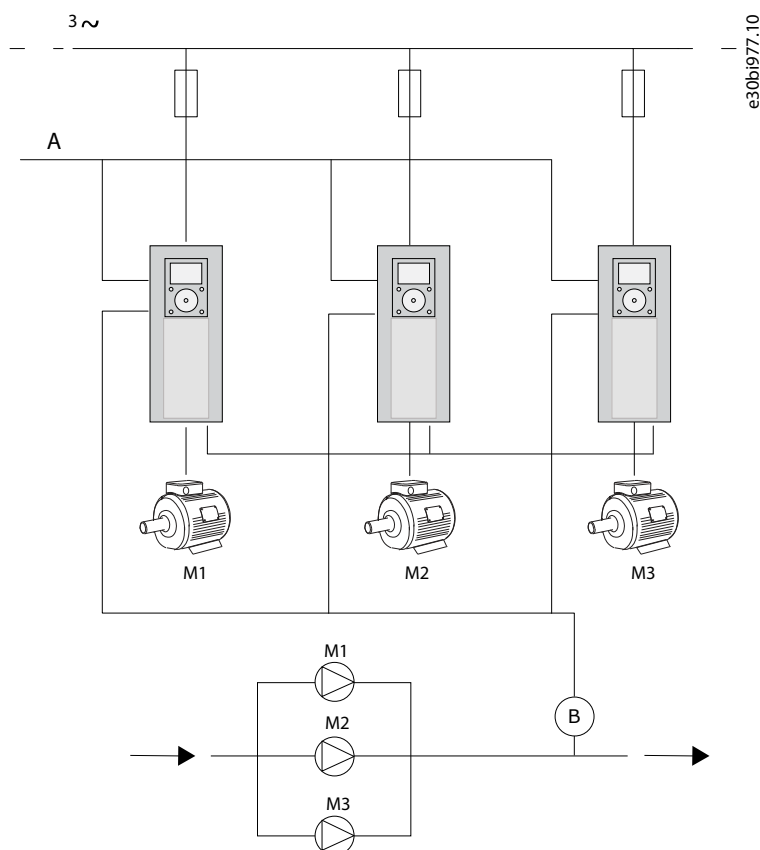


Illustration 15: Multi-pump (Multidrive) Configuration

A	Start / Stop
B	Pressure sensor

It is possible to control a process variable, the pressure, for example, by the control of the speed of the regulating motor and by the number of motors that operate. The internal PID controller in the drive of the regulating motor controls the speed, the start and stop of the motors.

The selected operation mode specifies the operation of the system. In the Multifollower mode, auxiliary motors follow the speed of the regulating motor.

Pump 1 controls and pumps 2 and 3 follow the speed of pump 1, as the curves A show.

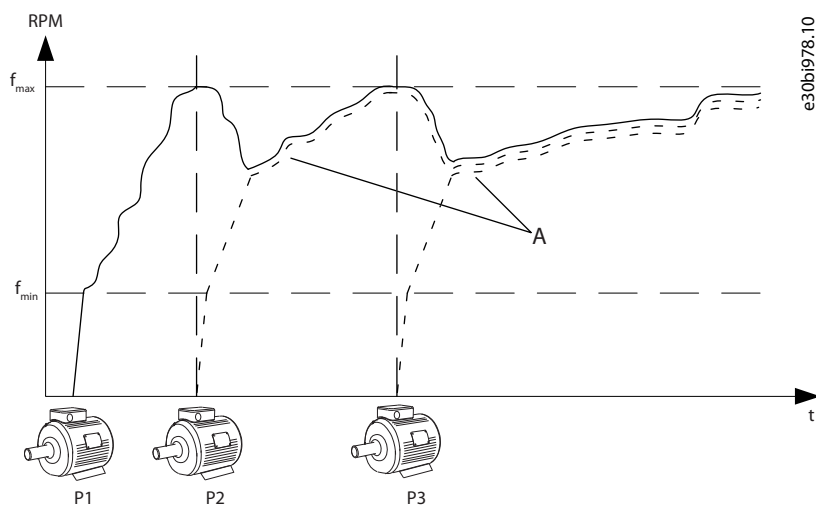


Illustration 16: Control in the Multifollower Mode

The figure [Illustration 17](#) shows an example of the Multimaster mode, where the speed of the regulating motor locks to the constant production speed B, when next motor starts. Curves A show the regulating of the pumps.

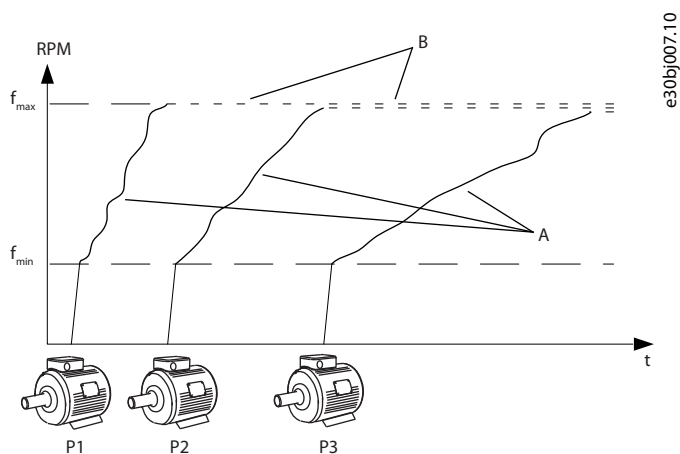


Illustration 17: Control in the Multimaster Mode

- | | |
|---|---|
| A | Curves A show the control of the pumps |
| B | The pumps are locked to the constant production frequency |

Autochange function (change of start order) makes the wear of the motors in the system more equal. Autochange function monitors the running hours and sets the start order of each motor. The motor that has the lowest running hours starts first and the motor that has the highest running hours starts last. It is possible to configure the autochange to start based on the autochange interval time or on the internal real-time clock of the drive (an RTC battery needed).

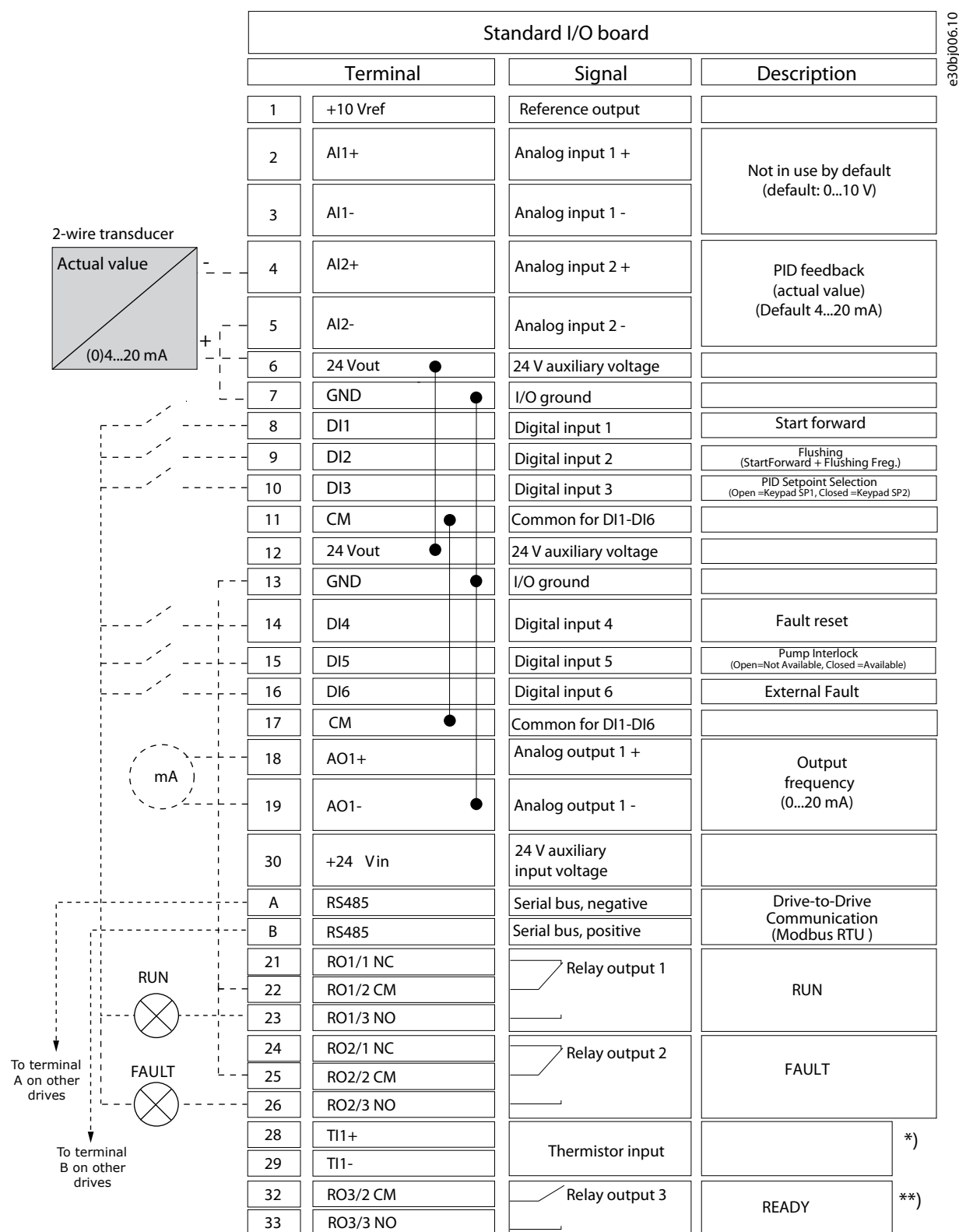


Illustration 18: The Default Control Connections of the Multi-pump (Multidrive) Application

*) Available only in VACON® 100 X. **) For the DIP switch configurations in VACON® 100 X, see the VACON® 100 X Installation manual.

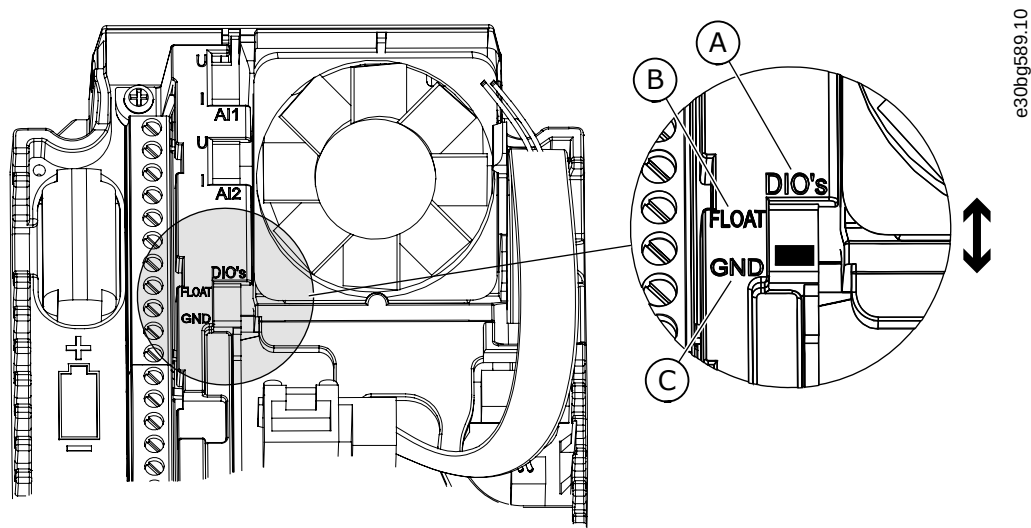


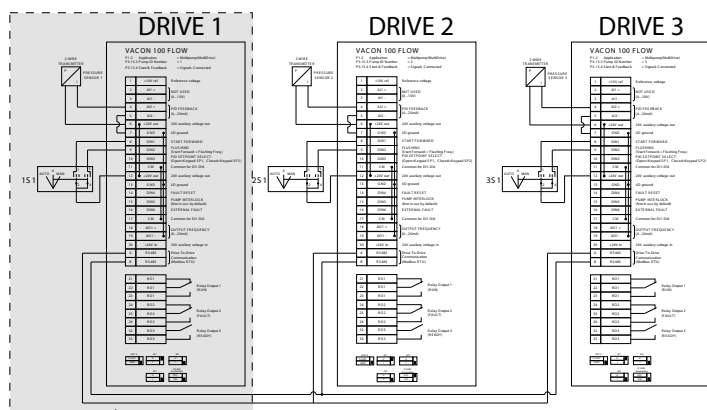
Illustration 19: The DIP Switch

A	Digital input DIP switch	C	Connected to GND (Default)
B	Floating		

Each drive has a pressure sensor. When the redundancy level is high, the drive and the pressure sensors are redundant.

- If there is a drive failure, the next drive starts to operate as master.
- If there is a sensor failure, the next drive (that has a separate sensor) starts to operate as master.

An individual switch that has an auto, off and man setting controls each drive.



e30bi990.10

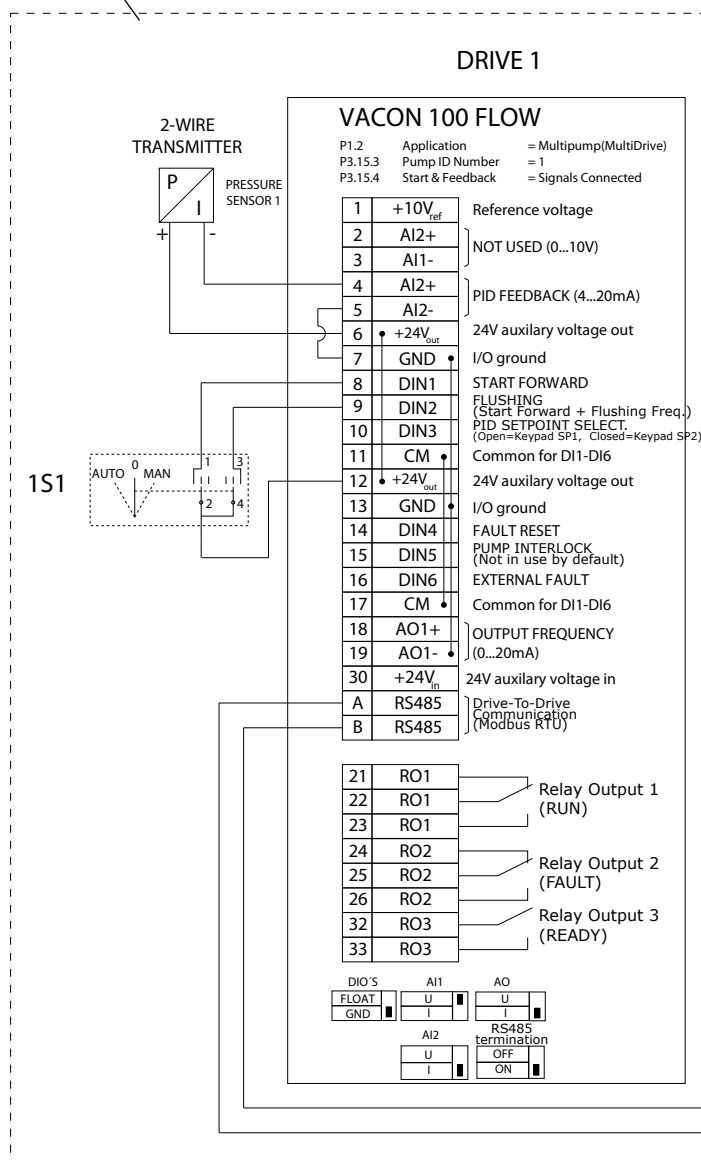
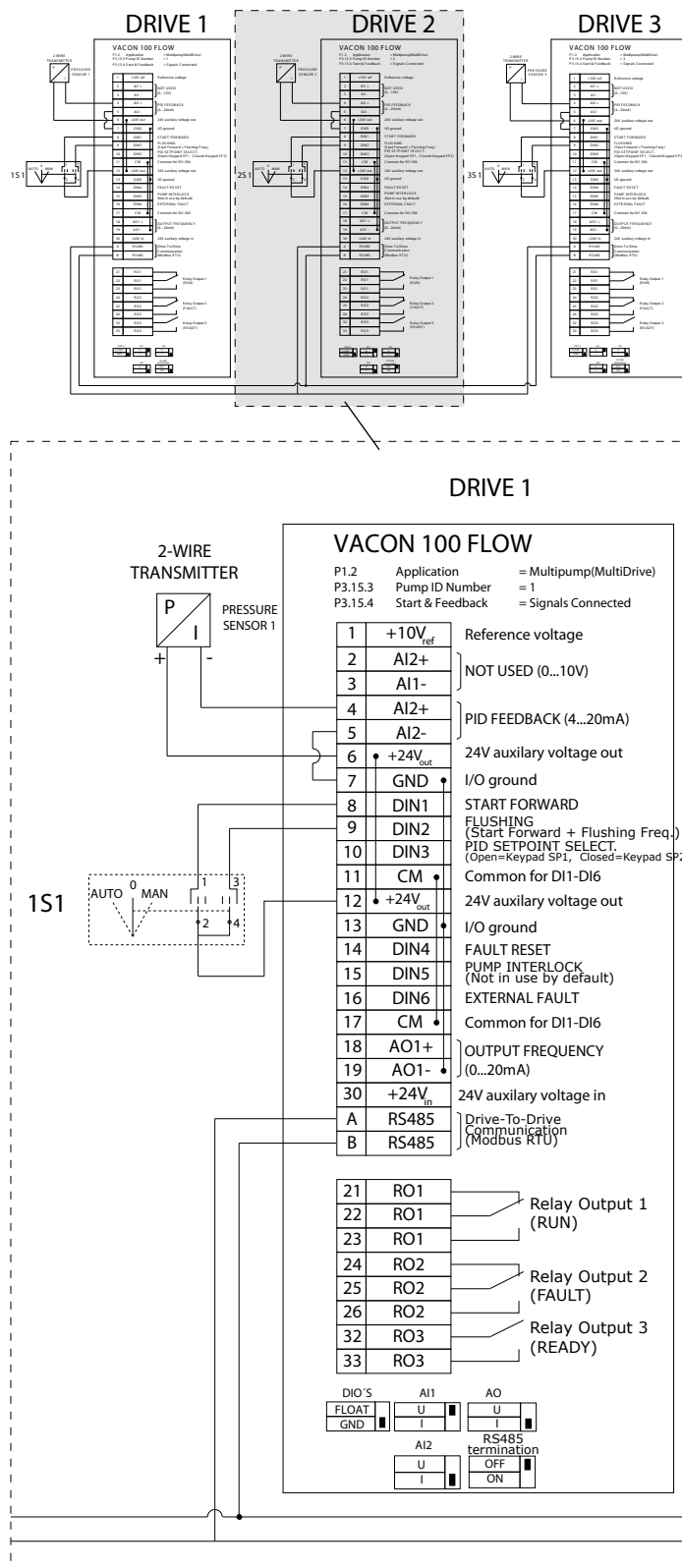


Illustration 20: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 1A



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Illustration 21: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 1B

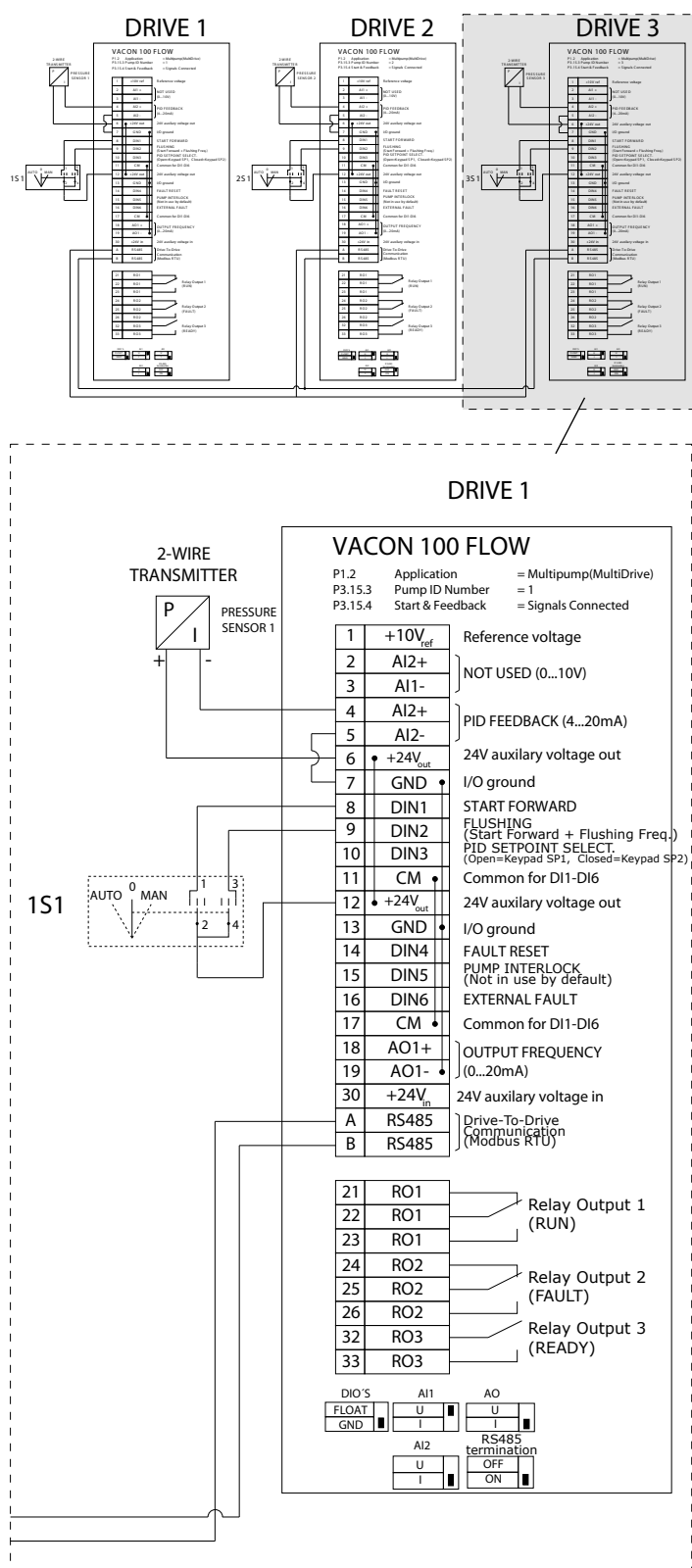
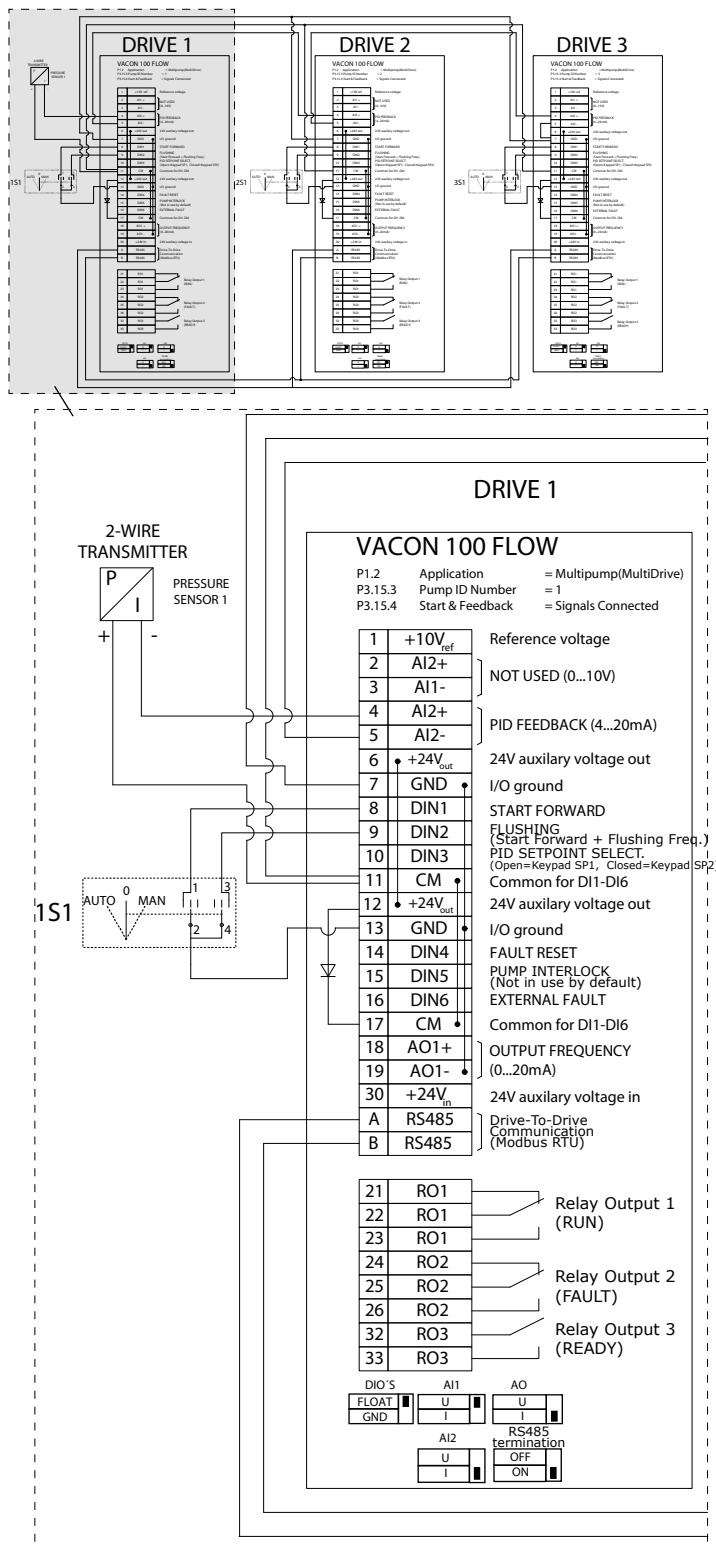


Illustration 22: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 1C

1 sensor is connected to all the drives. The redundancy level of the system is low because only the drives are redundant.

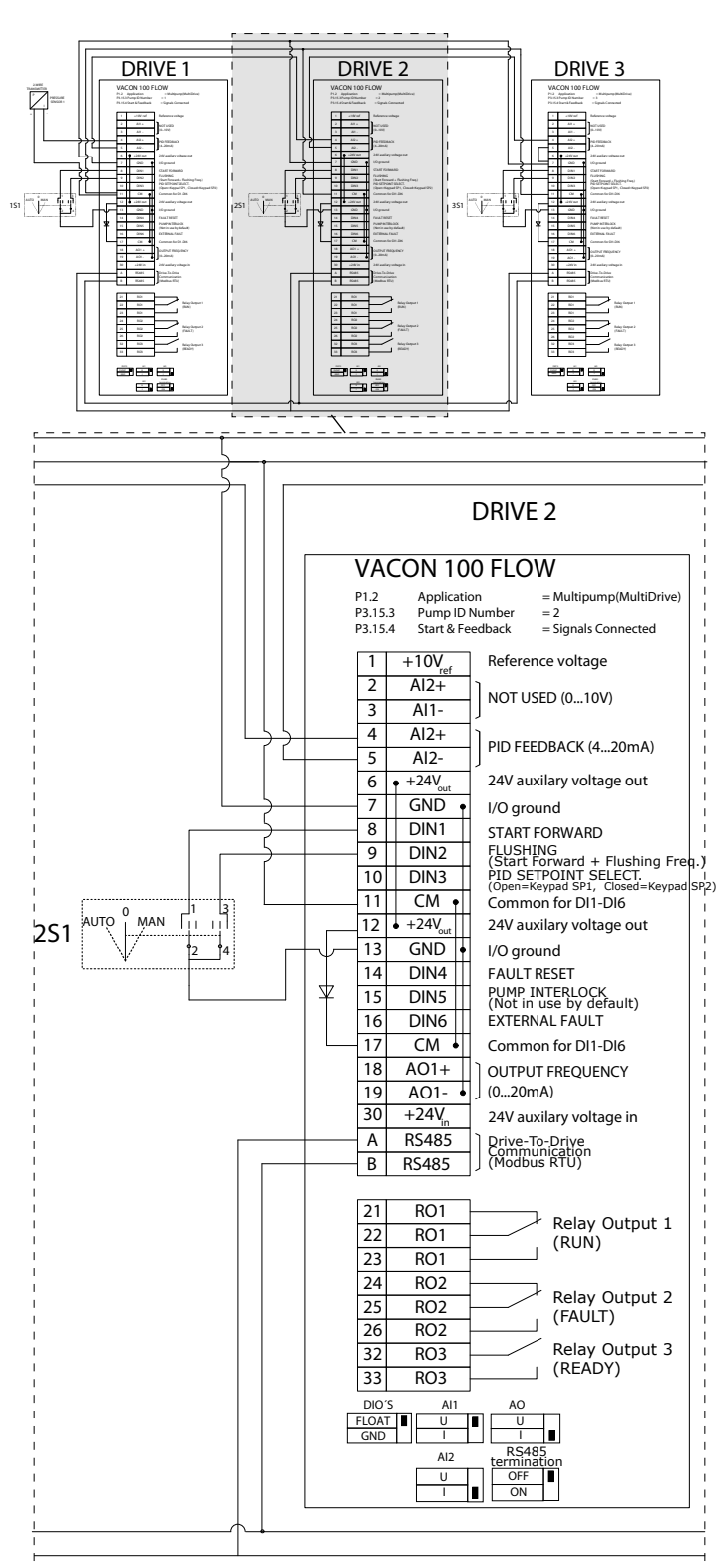
- If there is a drive failure, the next drive starts to operate as master.
- If there is a sensor failure, the system stops.

An individual switch that has an auto, off and man setting controls each drive. Terminal 17 connects +24 V between the drive 1 and 2. External diodes are connected between terminals 1 and 2. The digital input signals use negative logic (ON = OV).



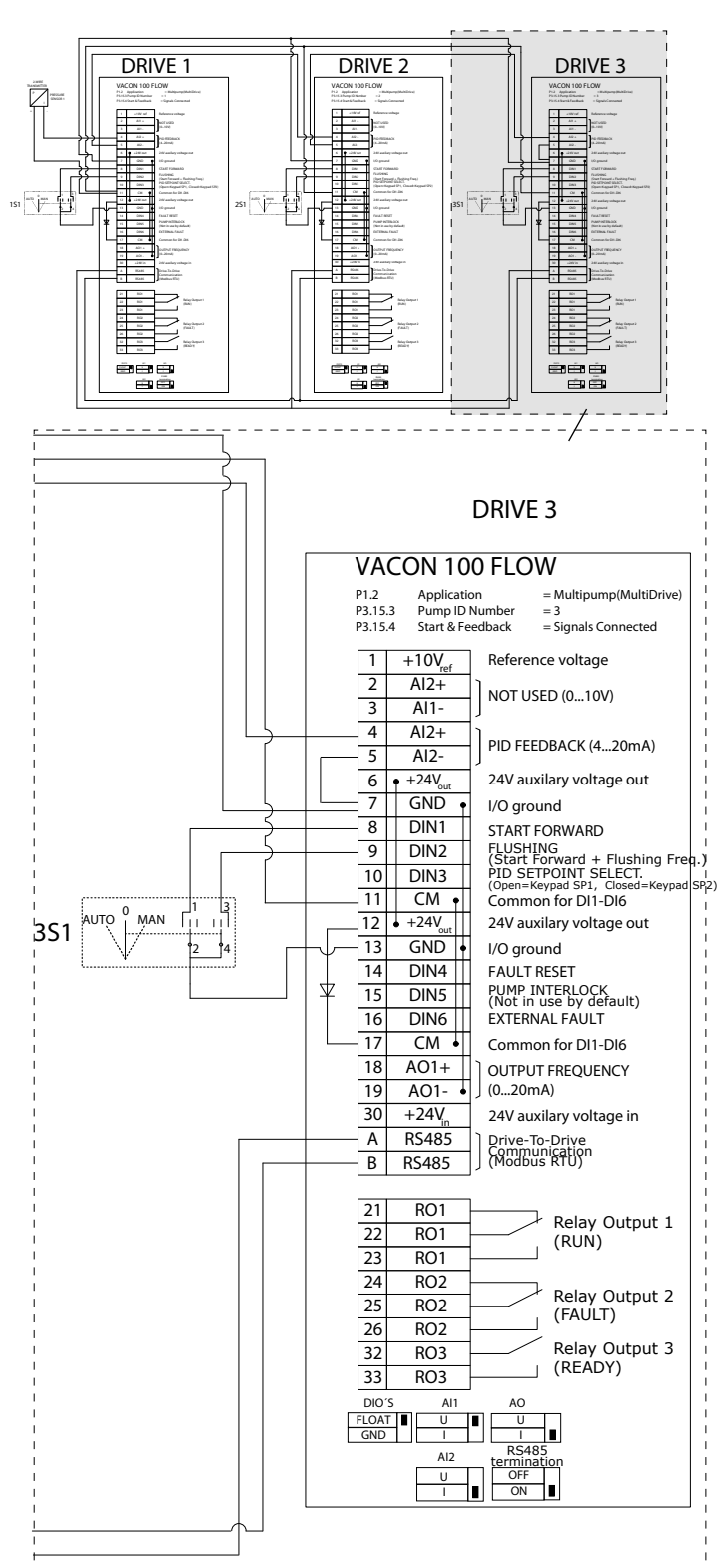
e30bi993.10

Illustration 23: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 2A



e30bi994.10

Illustration 24: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 2B



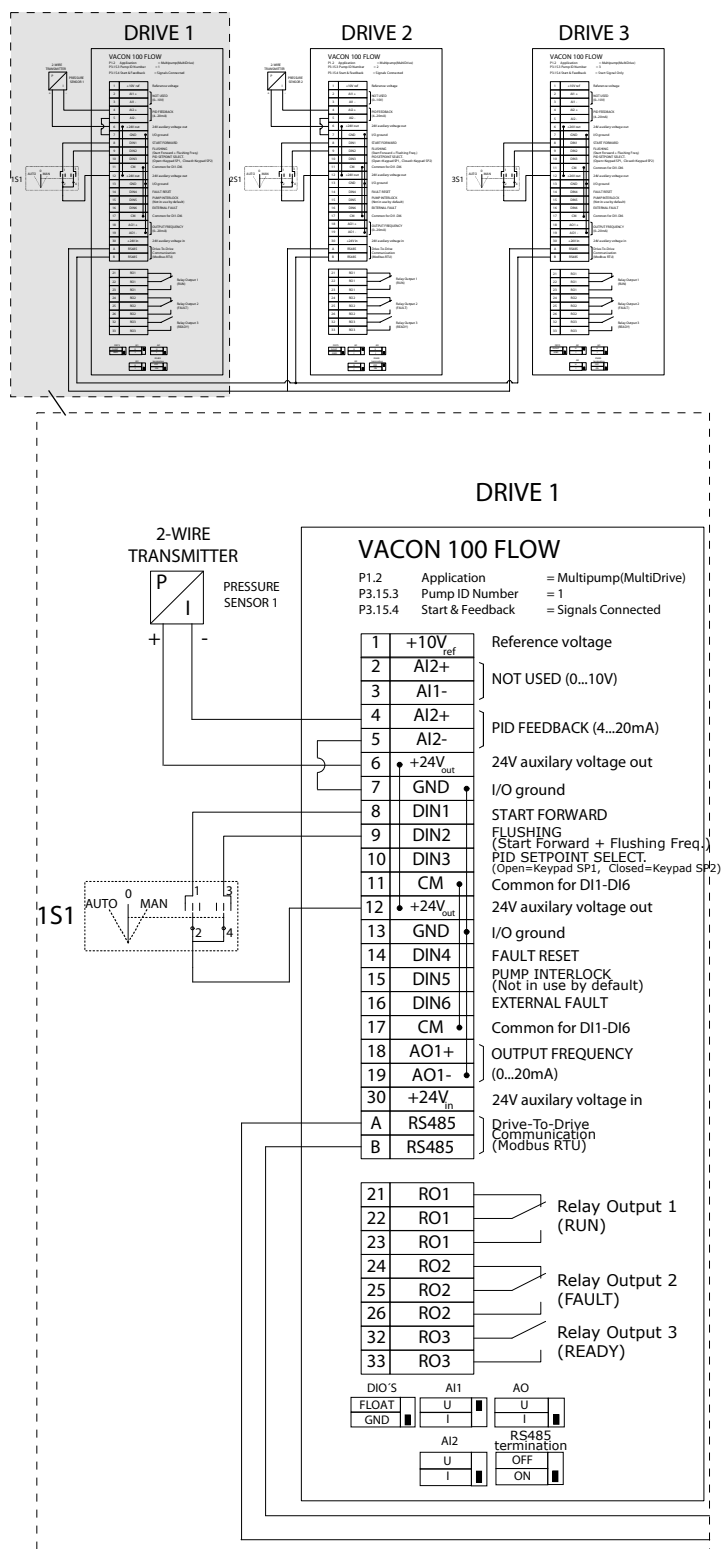
e30b1995.10

Illustration 25: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 2C

2 drives have individual pressure sensors. The redundancy level of the system is medium because the drives and the pressure sensors are duplicated.

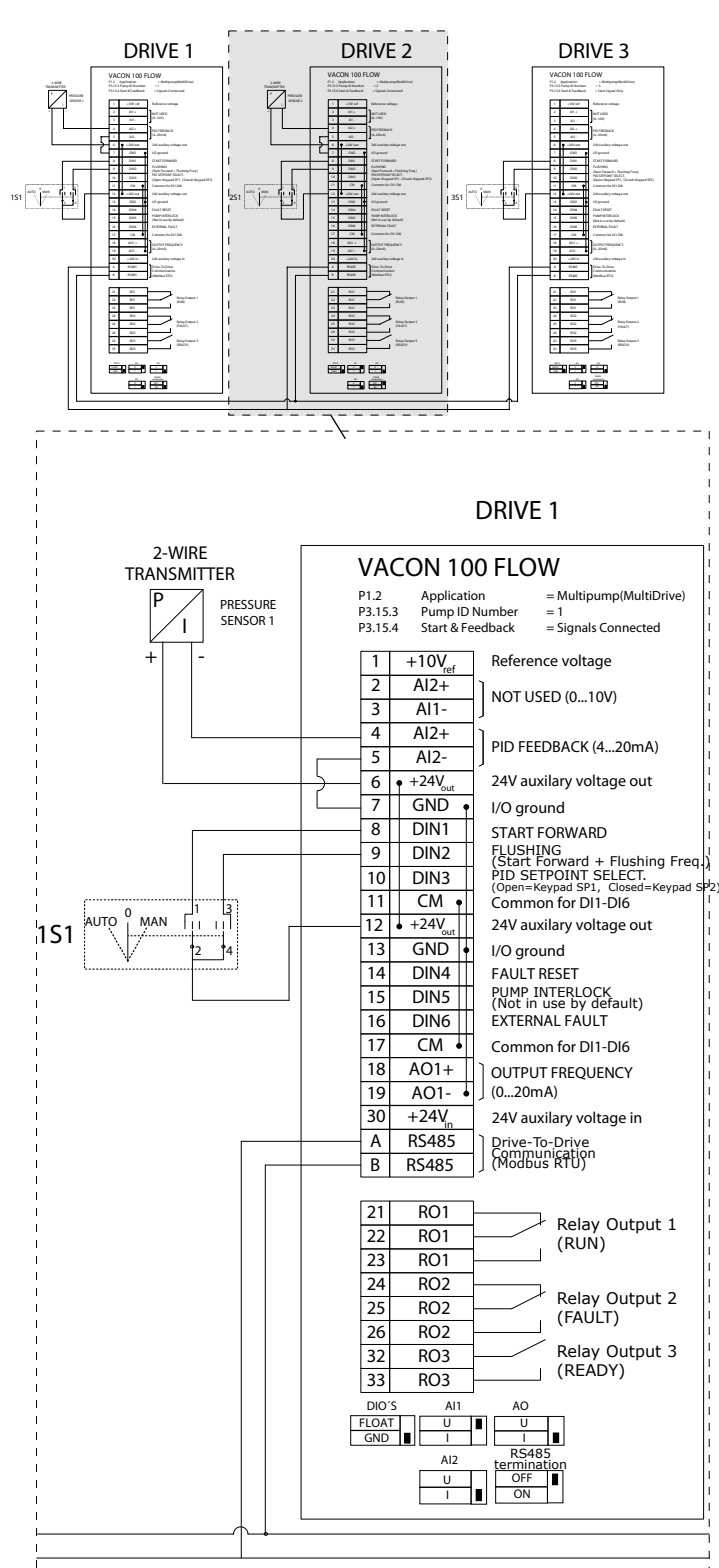
- If there is a drive failure, the second drive starts to operate as master.
- If there is a sensor failure, the second drive (that has a separate sensor) starts to operate as master.

An individual switch that has an auto, off and man setting controls each drive.



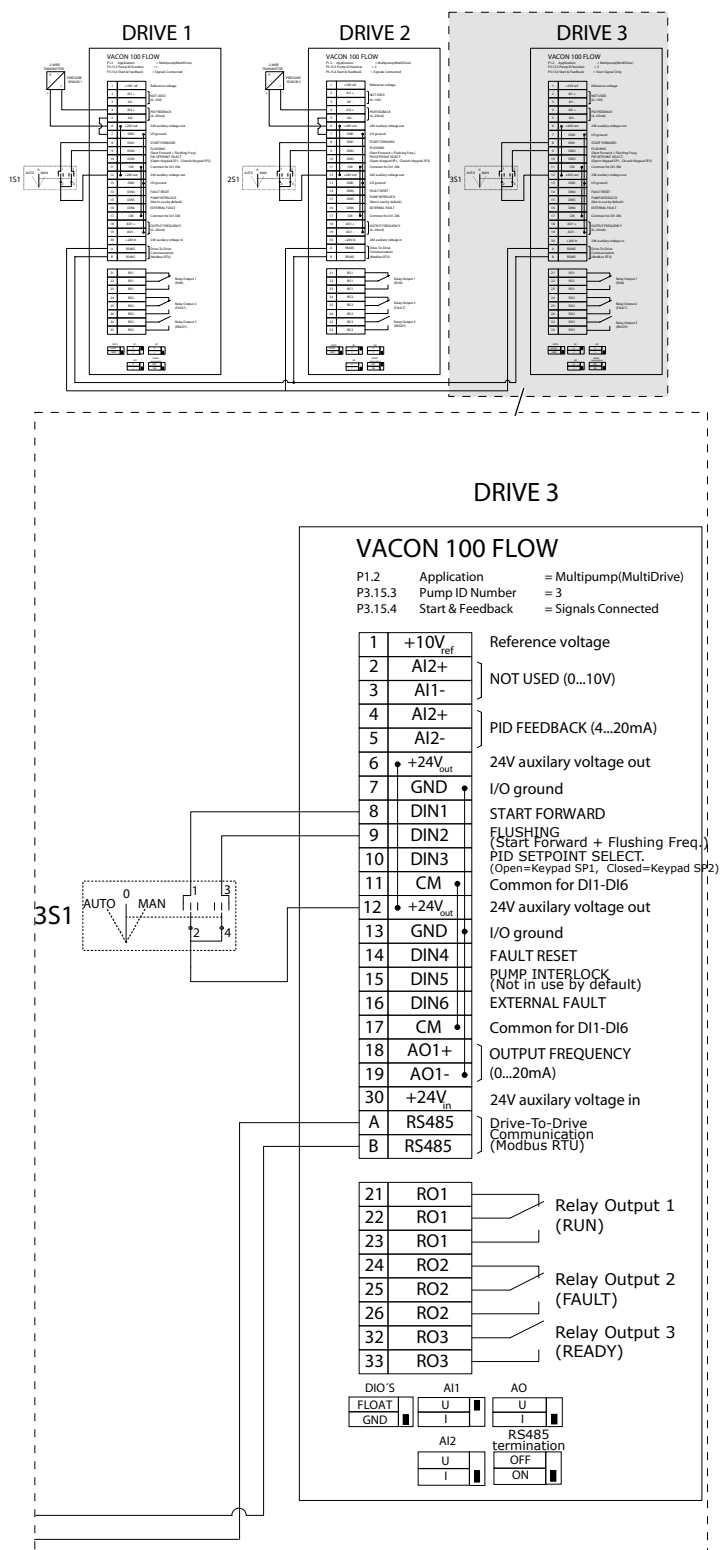
e30b1996.10

Illustration 26: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 3A



e30b997.10

Illustration 27: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 3B



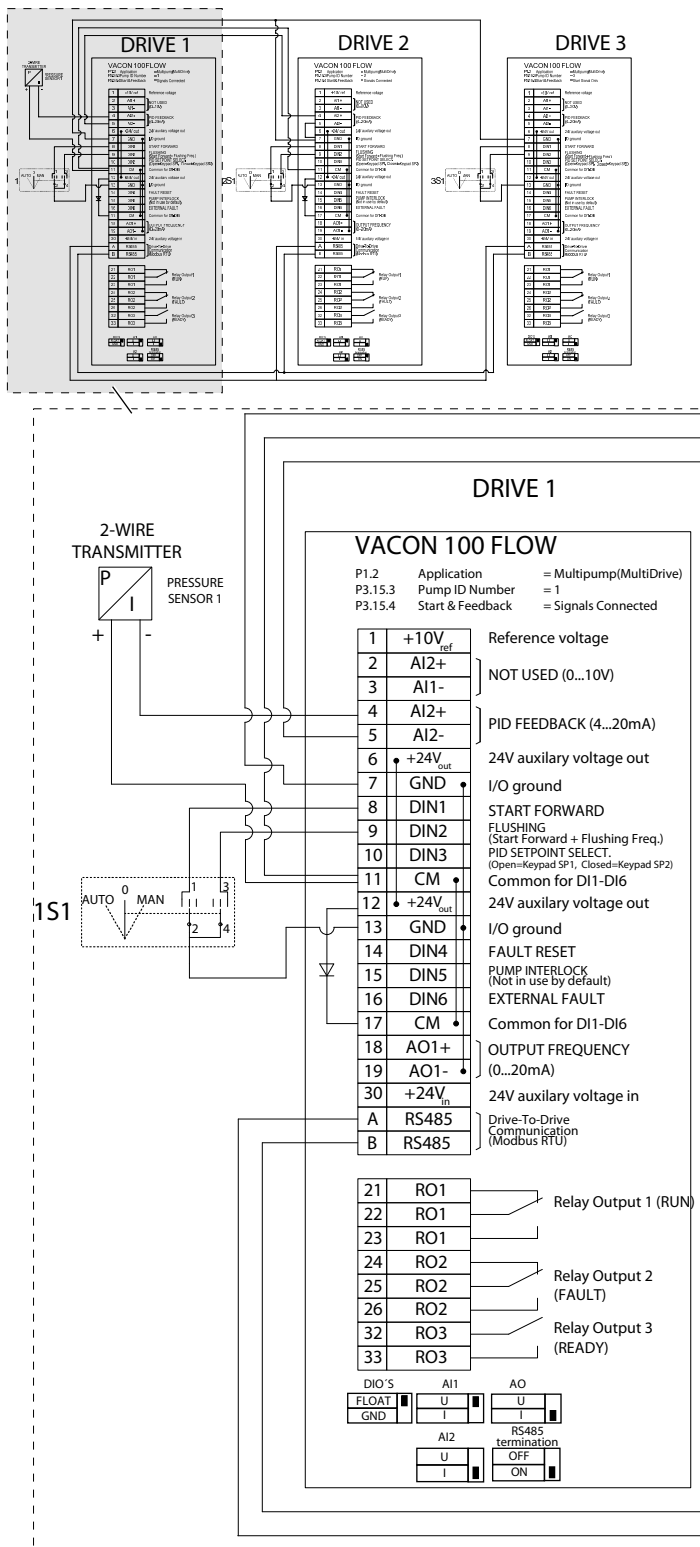
e30bi998.10

Illustration 28: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 3C

1 common pressure sensor is connected to 2 drives. The redundancy level of the system is low because only the drives are redundant.

- If there is a drive failure, the second drive starts to operate as master.
- If there is a sensor failure, the system stops.

An individual switch that has an auto, off and man setting controls each drive. Terminal 17 connects +24V between the drive 1 and 2. External diodes are connected between terminals 1 and 2. The digital input signals use negative logic (ON = OV).



e30bi999.10

Illustration 29: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 4A

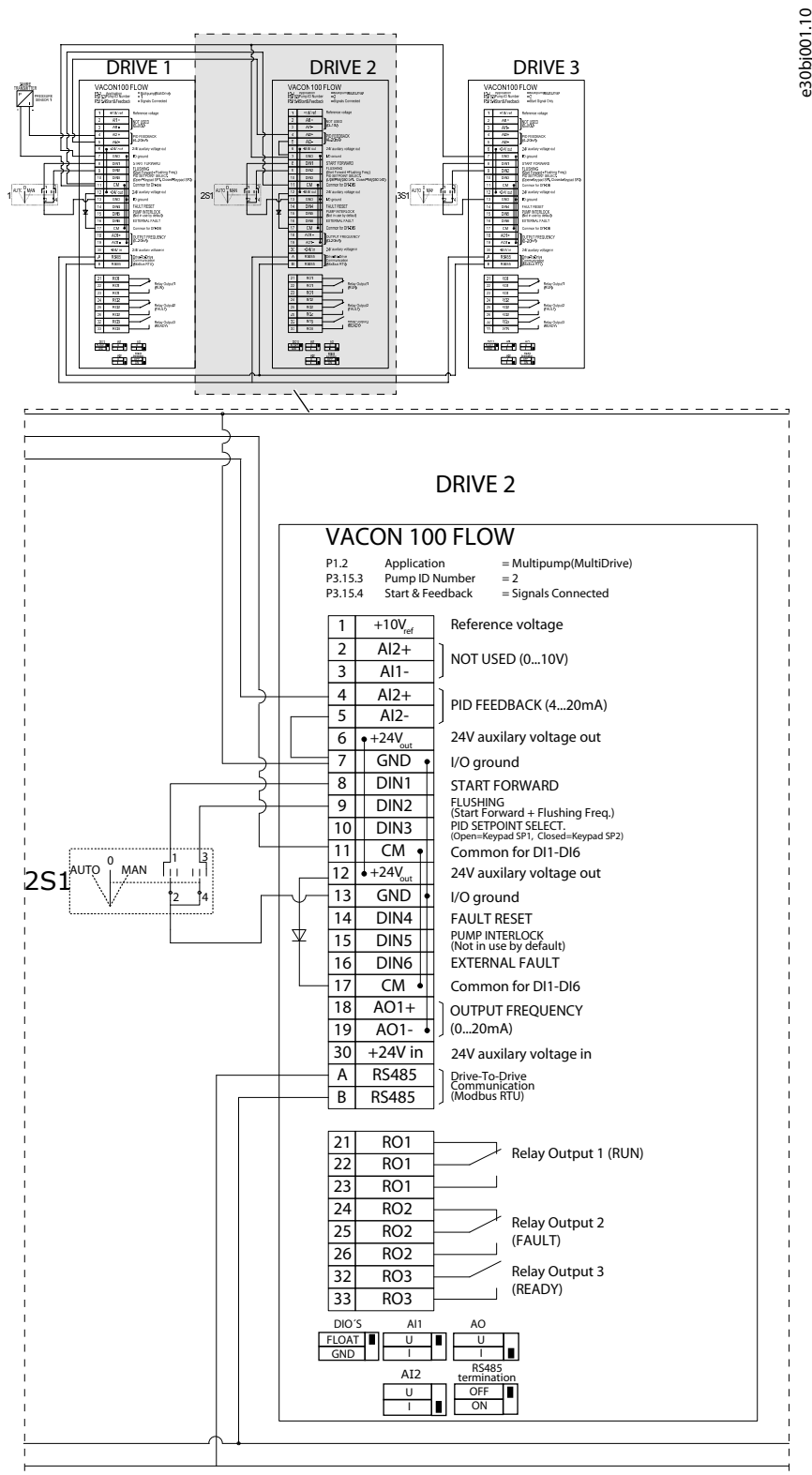
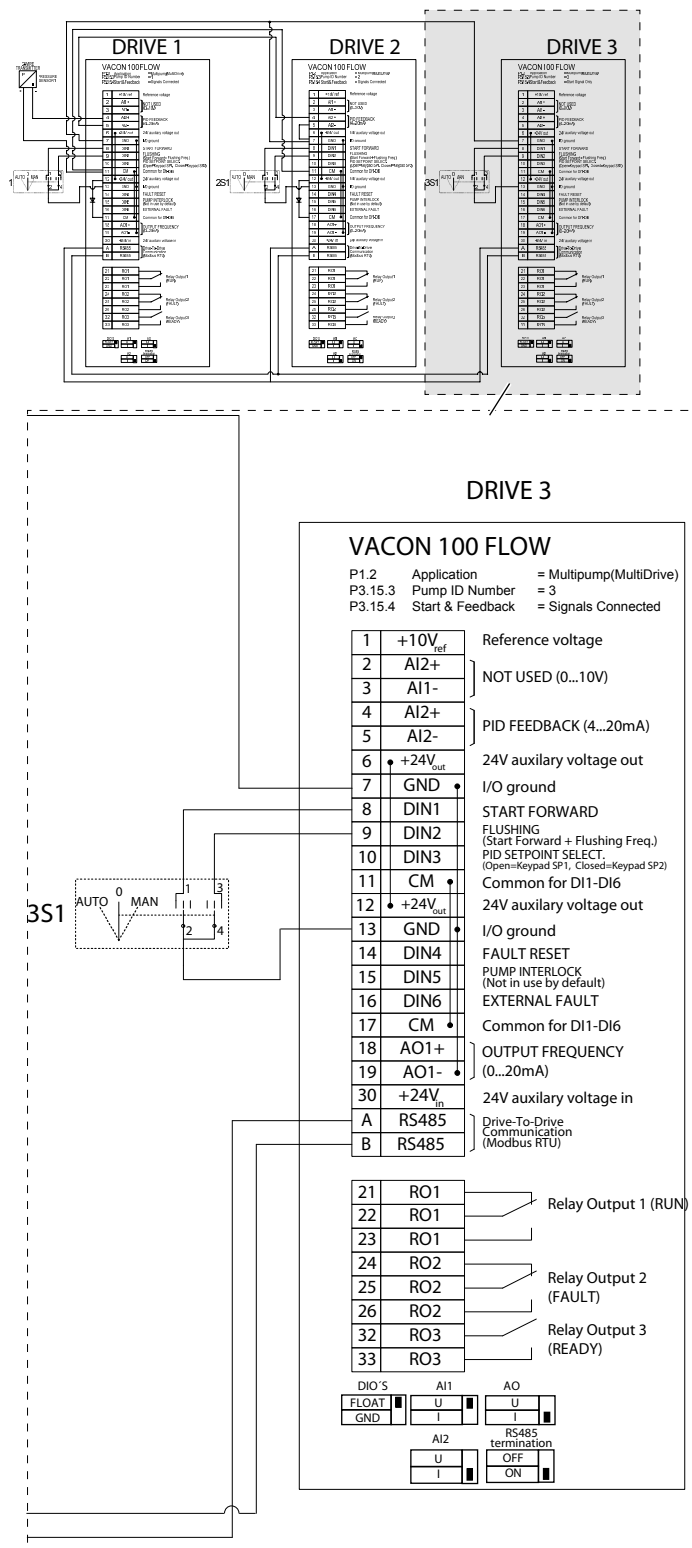


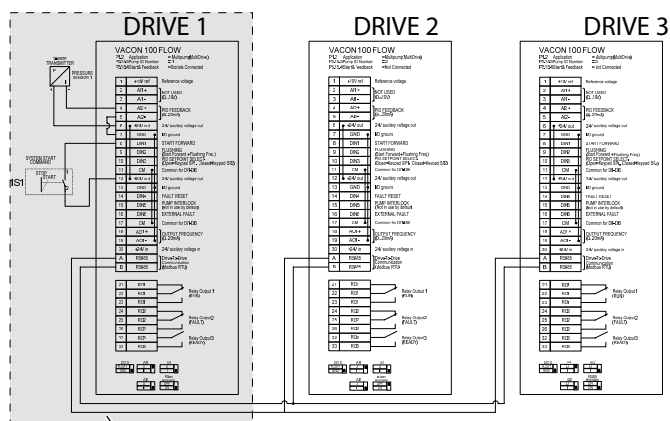
Illustration 30: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 4B



e30bj002.10

Illustration 31: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 4C

1 pressure sensor is connected to the first drive. The system is not redundant, because the system stops if there is a drive or sensor failure.



e30bj003.10

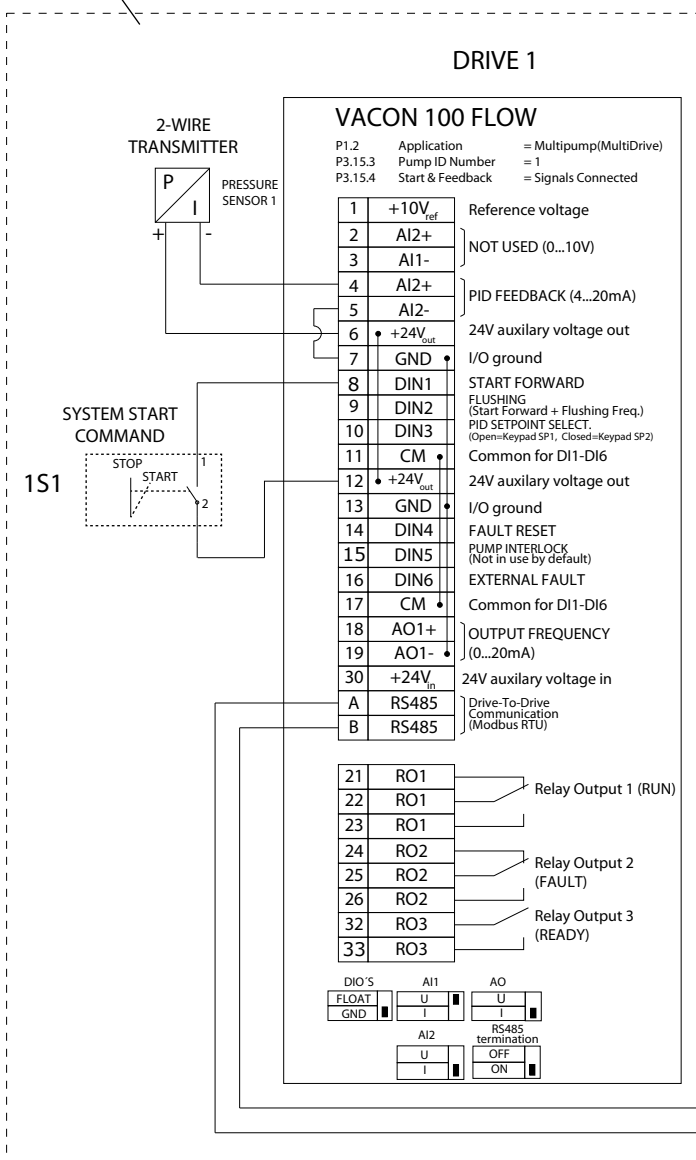
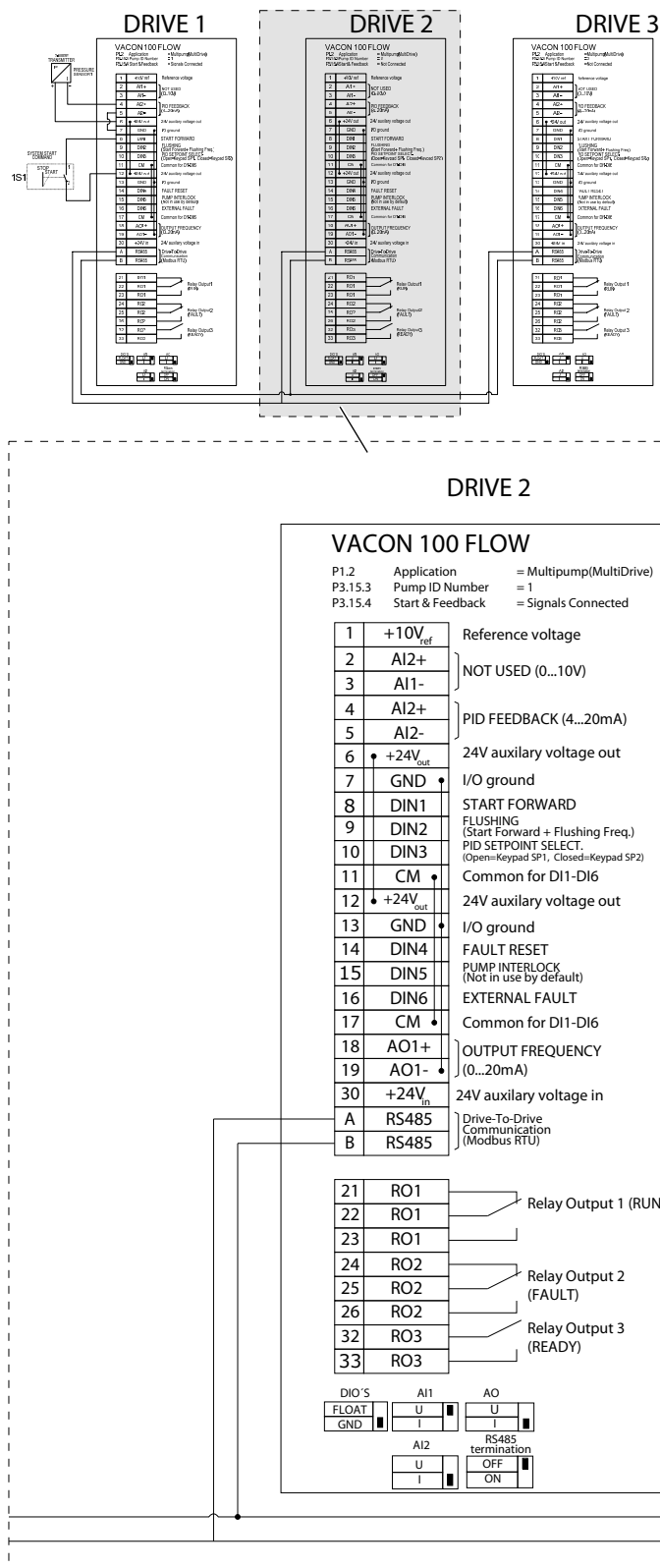


Illustration 32: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 5A



e30bj004.10

Illustration 33: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 5B

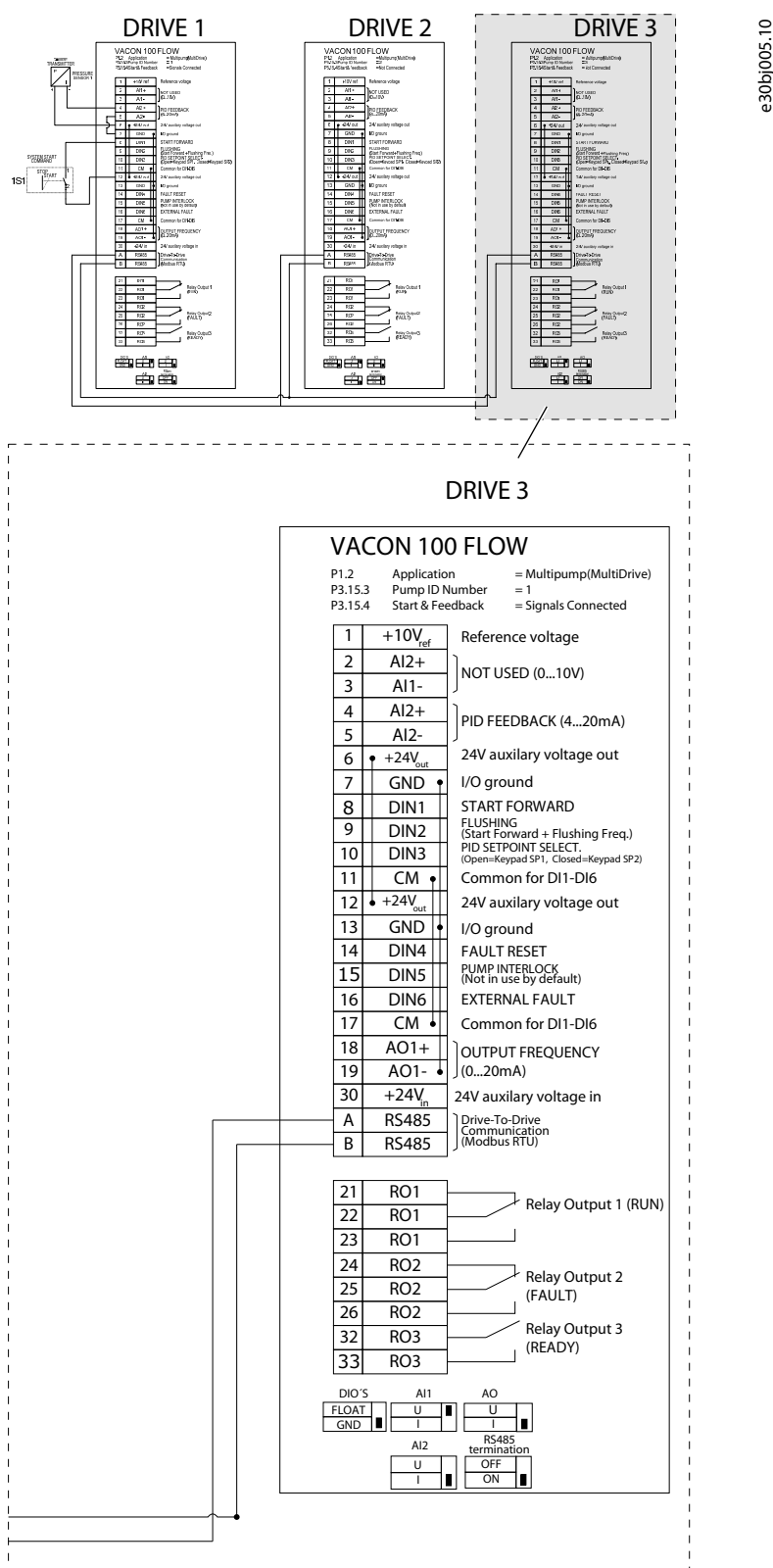


Illustration 34: Electric Wiring Diagram of the Multi-pump (Multidrive) System, Example 5C

See information on the Startup, and Fire mode wizards in [4.2.1 Wizards in Menu 1.1](#).

Table 9: M1 Quick Setup

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.2	Application	0	4		2	212	0 = Standard 1 = HVAC 2 = PID Control 3 = Multi-pump (single drive) 4 = Multi-pump (multidrive)
1.3	Minimum Frequency Reference	0.00	P1.4	Hz	0.0	101	
1.4	Maximum Frequency Reference	P1.3	320.0	Hz	50.0/60.0	102	
1.5	Acceleration Time 1	0.1	3000.0	s	5.0	103	
1.6	Deceleration Time 1	0.1	3000.0	s	5.0	104	
1.7	Motor Current Limit	I _H *0.1	I _S	A	Varies	107	
1.8	Motor Type	0	2		0	650	0 = Induction Motor 1 = Permanent Magnet Motor 2 = Reluctance Motor
1.9	Motor Nominal Voltage	Varies	Varies	V	Varies	110	Find this value U _n on the nameplate of the motor. NOTE! Find out if the motor connection is Delta or Star.
1.10	Motor Nominal Frequency	8.0	320.0	Hz	50/60	111	Find this value f _n on the nameplate of the motor.
1.11	Motor Nominal Speed	24	19200	RPM	Varies	112	Find this value n _n on the nameplate of the motor.
1.12	Motor Nominal Current	I _H * 0.1	I _S	A	Varies	113	Find this value I _n on the nameplate of the motor.
1.13	Motor Cos Phi (Power Factor)	0.30	1.00		Varies	120	Find this value on the nameplate of the motor.
1.14	Energy Optimization	0	1		0	666	0 = Disabled 1 = Enabled
1.15	Identification	0	2		0	631	0 = No action 1 = At standstill 2 = With rotation
1.16	Start Function	0	1		0	505	0 = Ramping 1 = Flying Start
1.17	Stop Function	0	1		0	506	0 = Coasting 1 = Ramping
1.18	Automatic Reset	0	1		0	731	0 = Disabled 1 = Enabled

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.19	Response to External Fault	0	3		2	701	0 = No action 1 = Alarm 2 = Fault (Stop according to stop mode) 3 = Fault (Stop by coasting)
1.20	Response to AI Low Fault	0	5		0	700	0 = No action 1 = Alarm 2 = Alarm+preset fault frequency (P3.9.1.13) 3 = Alarm + previous frequency 4 = Fault (Stop according to stop mode) 5 = Fault (Stop by coasting)
1.21	Remote Control Place	0	1		0	172	0 = I/O control 1 = Fieldbus control
1.22	I/O Control Reference A Selection	1	20		6	117	1 = Preset Frequency 0 2 = Keypad reference 3 = Fieldbus 4 = AI1 5 = AI2 6 = AI1+AI2 7 = PID reference 8 = Motor potentiometer 11 = Block Out.1 12 = Block Out.2 13 = Block Out.3 14 = Block Out.4 15 = Block Out.5 16 = Block Out.6 17 = Block Out.7 18 = Block Out.8 19 = Block Out.9 20 = Block Out.10
1.23	Keypad Control Reference Selection	1	20		1	121	See P1.22.
1.24	Fieldbus Control Reference Selection	1	20		2	122	See P1.22.
1.25	AI1 Signal Range	0	1		0	379	0= 0...10 V/0...20 mA 1= 2...10 V/4...20 mA
1.26	AI2 Signal Range	0	1		1	390	0= 0...10 V/0...20 mA 1= 2...10 V/4...20 mA
1.27	RO1 Function	0	74		2	11001	See P3.5.3.2.1.
1.28	RO2 Function	0	74		3	11004	See P3.5.3.2.1.
1.29	RO3 Function	0	74		1	11007	See P3.5.3.2.1.
1.30	AO1 Function	0	31		2	10050	See P3.5.4.1.1.

Table 10: M1.34 Multi-pump (Multidrive)

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.35.1	PID Gain	0.00	100.00	%	100.00	118	
1.35.2	PID Integration Time	0.00	600.00	s	1.00	119	
1.35.3	PID Derivation Time	0.00	100.00	s	0.00	132	
1.35.4	Process Unit Selection	1	44		1	1036	See P3.13.1.4
1.35.5	Process Unit Min	Varies	Varies		Varies	1033	
1.35.6	Process Unit Max	Varies	Varies		Varies	1034	
1.35.7	Feedback 1 Source Selection	0	30		2	334	See P3.13.3.3
1.35.8	Setpoint 1 Source Selection	0	32		1	332	See P3.13.2.6
1.35.9	Keypad Setpoint 1	Varies	Varies	Varies	0	167	
1.35.10	SP1 Sleep Frequency Limit	0.0	320.0	Hz	0.0	1016	
1.35.11	SP1 Sleep Delay	0	3000	s	0	1017	0 = Not used
1.35.12	SP1 Wake Up Level	Varies	Varies	Varies	Varies	1018	0 = Not used
1.35.13	Multi-pump Mode	0	2		0	1785	0= Single drive 1= Multifollower 2= Multimaster
1.35.14	Number of Pumps	1	8		1	1001	
1.35.15	Pump ID Number	1	8		1	1500	
1.35.16	Start and Feedback Signals	0	2		1	1782	0 = Not connected 1 = Only Start Signal Connected 2 = Both Signals Connected
1.35.17	Pump Interlocking	0	1		1	1032	0 = Disabled 1 = Enabled
1.35.18	Autochange	0	2		1	1027	0 = Disabled 1 = Enabled (interval) 2 = Enabled (weekdays)
1.35.19	Autochanged Pump	0	1		1	1028	0 = Auxiliary Pump 1 = All Pumps
1.35.20	Autochange Interval	0.0	3000.0	h	48.0	1029	
1.35.21	Autochange Days	0	127			1786	
1.35.22	Autochange Time of Day			Time		1787	
1.35.23	Autochange: Frequency Limit	0.00	P3.3.1.2	Hz	25:00	1031	
1.35.24	Autochange: Pump Limit	1	6			1030	

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.35.25	Bandwidth	0	100	%	10	1097	
1.35.26	Bandwidth Delay	0	3600	s	10	1098	
1.35.27	Constant Production Speed	0	100	%	100	1513	
1.35.28	Pump 1 Interlock				DigIN Slot0.1	426	OPEN = Not active CLOSED = Active
1.35.29	Flushing Reference	Maximum reference	Maximum reference	Hz	50.00	1239	

4 Wizards

4.1 Application Wizards

4.1.1 Standard Application Wizard

The application wizard helps to set the basic parameters that are related to the application.

To start the Standard application wizard, set the value Standard to parameter P1.2 Application (ID 212) in the keypad.

If the application wizard is started from the Start-up wizard, the wizard goes directly to step 11.

See the range of values of the parameters in [6.1 Group 3.1: Motor Settings](#), [6.3 Group 3.3: References](#), [6.4 Group 3.4: Ramps and Brakes Setup](#).

Procedure

1. Set a value for P3.1.2.2 Motor Type (so that it agrees with the nameplate of the motor).

➡ If Motor Type is set to Induction Motor, also step 6 is shown. If the selection is PM Motor, the value of parameter P3.1.1.5 Motor Cos Phi is set to 1.00, and the wizard skips step 6.

2. Set a value for P3.1.1.1 Motor Nominal Voltage (so that it agrees with the nameplate of the motor).
3. Set a value for P3.1.1.2 Motor Nominal Frequency (so that it agrees with the nameplate of the motor).
4. Set a value for P3.1.1.3 Motor Nominal Speed (so that it agrees with the nameplate of the motor).
5. Set a value for P3.1.1.4 Motor Nominal Current (so that it agrees with the nameplate of the motor).
6. Set a value for P3.1.1.5 Motor Cos Phi (so that it agrees with the nameplate of the motor).
7. Set a value for P3.3.1.1 Minimum Frequency Reference.
8. Set a value for P3.3.1.2 Maximum Frequency Reference.
9. Set a value for P3.4.1.2 Acceleration Time 1.
10. Set a value for P3.4.1.3 Deceleration Time 1.
11. Select the a control place (where to give the start and stop commands, and the frequency reference of the drive).

➡ The Standard application wizard is completed.

4.1.2 HVAC Application Wizard

The application wizard helps to set the basic parameters that are related to the application.

To start the HVAC application wizard, set the value Standard to parameter P1.2 Application (ID 212) in the keypad.

See the range of values of the parameters in [6.2 Group 3.2: Start/Stop Setup](#), [6.1 Group 3.1: Motor Settings](#), [6.3 Group 3.3: References](#), [6.4 Group 3.4: Ramps and Brakes Setup](#).

Procedure

1. Select the type or process (or application) to control.
 - Compressor
 - Fan
 - Pump
 - Other

Some parameters have preset values specified by the selection. See the parameters and their values in the following table.

Table 11: Preset Values of Parameters

Index	Parameter	Process type		
		Pump	Fan	Compressor
P3.1.4.1	U/f Ratio	Linear	Squared	Linear
P3.2.4	Start Function	Ramping	Flying start	Ramping
P3.2.5	Stop Function	Ramping	Coasting	Ramping
P3.4.1.2	Acceleration Time	5.0 s	30.0 s	3.0 s
P3.4.1.3	Deceleration Time	5.0 s	30.0 s	3.0 s

➡ If process is set to Compressor, also step 2 is shown. If the selection is anything else, the wizard skips step 2. If process is set to Other, also steps 11 and 12 are shown.

2. Set a value for P3.2.11 Restart Delay.
3. Set a value for P3.1.2.2 Motor Type (so that it agrees with the nameplate of the motor).

➡ If Motor Type is set to Induction Motor, also step 8 is shown. If the selection is PM Motor, the value of parameter P3.1.1.5 Motor Cos Phi is set to 1.00, and the wizard skips step 8.

4. Set a value for P3.1.1.1 Motor Nominal Voltage (so that it agrees with the nameplate of the motor).
5. Set a value for P3.1.1.2 Motor Nominal Frequency (so that it agrees with the nameplate of the motor).
6. Set a value for P3.1.1.3 Motor Nominal Speed (so that it agrees with the nameplate of the motor).
7. Set a value for P3.1.1.4 Motor Nominal Current (so that it agrees with the nameplate of the motor).
8. Set a value for P3.1.1.5 Motor Cos Phi (so that it agrees with the nameplate of the motor).
9. Set a value for P3.3.1.1 Minimum Frequency Reference.
10. Set a value for P3.3.1.2 Maximum Frequency Reference.
11. Set a value for P3.4.1.2 Acceleration Time 1.
12. Set a value for P3.4.1.3 Deceleration Time 1.

➡ The wizard goes to steps specified by the application.

13. Select the control place (where to give the start and stop commands, and the frequency reference of the drive).

➡ The HVAC application wizard is completed.

4.1.3 PID Control Application Wizard

The application wizard helps to set the basic parameters that are related to the application.

To start the PID Control application wizard, set the value PID Control to parameter P1.2 Application (ID 212) in the keypad.

If the application wizard is started from the Start-up wizard, the wizard goes directly to step 11.

See the range of values of the parameters in [6.1 Group 3.1: Motor Settings](#), [6.3 Group 3.3: References](#), [6.4 Group 3.4: Ramps and Brakes Setup](#), [6.13 Group 3.13: PID Controller](#).

Procedure

1. Set a value for P3.1.2.2 Motor Type (so that it agrees with the nameplate of the motor).

➡ If Motor Type is set to Induction Motor, also step 6 is shown. If the selection is PM Motor, the value of parameter P3.1.1.5 Motor Cos Phi is set to 1.00, and the wizard skips step 6.

2. Set a value for P3.1.1.1 Motor Nominal Voltage (so that it agrees with the nameplate of the motor).
3. Set a value for P3.1.1.2 Motor Nominal Frequency (so that it agrees with the nameplate of the motor).
4. Set a value for P3.1.1.3 Motor Nominal Speed (so that it agrees with the nameplate of the motor).
5. Set a value for P3.1.1.4 Motor Nominal Current (so that it agrees with the nameplate of the motor).
6. Set a value for P3.1.1.5 Motor Cos Phi (so that it agrees with the nameplate of the motor).
7. Set a value for P3.3.1.1 Minimum Frequency Reference.
8. Set a value for P3.3.1.2 Maximum Frequency Reference.
9. Set a value for P3.4.1.2 Acceleration Time 1.
10. Set a value for P3.4.1.3 Deceleration Time 1.
11. Select a control place (where to give the start and stop commands, and the frequency reference of the drive).
12. Set a value for P3.13.1.4 Process Unit Selection.

➡ If the selection is other than %, the next steps are shown. If the selection is %, the wizard goes directly to step 16.

13. Set a value for P3.13.1.5 Process Unit Min.

The range depends on the selection in step 12.

14. Set a value for P3.13.1.6 Process Unit Max.

The range depends on the selection in step 12.

15. Set a value for P3.13.1.7 Process Unit Decimals.

16. Set a value for P3.13.3.3 Feedback 1 Source Selection.

See the table Feedback settings in chapter [6.13 Group 3.13: PID Controller](#).

➡ If analog input signal is selected, step 18 is shown. With other selections, the wizard goes to step 19.

17. Set the signal range of the analog input.
18. Set a value for P3.13.1.8 Error Inversion.
19. Set a value for P3.13.2.6 Setpoint Source Selection.

See table Setpoints in chapter [6.13 Group 3.13: PID Controller](#).

➡ If analog input signal is selected, step 21 is shown. With other selections, the wizard goes to step 23. If Keypad Setpoint 1 or Keypad Setpoint 2 are set for the value, the wizard goes directly to step 22.

20. Set the signal range of the analog input.
21. Set a value for P3.13.2.1 (Keypad Setpoint 1) and P3.13.2.2 (Keypad Setpoint 2).

Depends on the range set in step 20.

22. Using the sleep function.

➡ If value Yes is set for step 22, the next 3 steps are shown. If the value No is selected, the wizard is completed.

23. Set a value for P3.13.5.1 SP1 Sleep Frequency Limit.
24. Set a value for P3.13.5.2 SP1 Sleep Delay.
25. Set a value for P3.13.5.3 SP1 Wake-up Level.

The range depends on the set process unit.

➡ The PID Control application wizard is completed.

4.1.4 Multi-pump (Single Drive) Application Wizard

The application wizard helps to set the basic parameters that are related to the application.

To start the Multi-pump (single drive application wizard, set the value PID Control to parameter P1.2 Application (ID 212) in the keypad.

If the application wizard is started from the Start-up wizard, the wizard goes directly to step 11.

See the range of values of the parameters in [6.1 Group 3.1: Motor Settings](#), [6.3 Group 3.3: References](#), [6.4 Group 3.4: Ramps and Brakes Setup](#), [6.13 Group 3.13: PID Controller](#).

Procedure

1. Set a value for P3.1.2.2 Motor Type (so that it agrees with the nameplate of the motor).

➡ If Motor Type is set to Induction Motor, also step 6 is shown. If the selection is PM Motor, the value of parameter P3.1.1.5 Motor Cos Phi is set to 1.00, and the wizard skips step 6.

2. Set a value for P3.1.1.1 Motor Nominal Voltage (so that it agrees with the nameplate of the motor).
3. Set a value for P3.1.1.2 Motor Nominal Frequency (so that it agrees with the nameplate of the motor).
4. Set a value for P3.1.1.3 Motor Nominal Speed (so that it agrees with the nameplate of the motor).
5. Set a value for P3.1.1.4 Motor Nominal Current (so that it agrees with the nameplate of the motor).
6. Set a value for P3.1.1.5 Motor Cos Phi (so that it agrees with the nameplate of the motor).
7. Set a value for P3.3.1.1 Minimum Frequency Reference.
8. Set a value for P3.3.1.2 Maximum Frequency Reference.
9. Set a value for P3.4.1.2 Acceleration Time 1.
10. Set a value for P3.4.1.3 Deceleration Time 1.
11. Select a control place (where to give the start and stop commands, and the frequency reference of the drive).
12. Set a value for P3.13.1.4 Process Unit Selection.

➡ If the selection is other than %, the next steps are shown. If the selection is %, the wizard goes directly to step 16.

13. Set a value for P3.13.1.5 Process Unit Min.

14. Set a value for P3.13.1.6 Process Unit Max.
15. Set a value for P3.13.1.7 Process Unit Decimals.
16. Set a value for P3.13.3.3 Feedback 1 Source Selection.

See the table Feedback settings in chapter [6.13 Group 3.13: PID Controller](#).

➡ If analog input signal is selected, step 17 is shown. With other selections, the wizard goes to step 18.

17. Set the signal range of the analog input.

See table Analog inputs in chapter [6.5 Group 3.5: I/O Configuration](#).

18. Set a value for P3.13.1.8 Error Inversion.
19. Set a value for P3.13.2.6 Setpoint Source Selection.

See table Setpoints in chapter [6.13 Group 3.13: PID Controller](#).

➡ If analog input signal is selected, step 20 is shown. With other selections, the wizard goes to step 21. If Keypad Setpoint 1 or Keypad Setpoint 2 is selected for the value, step 22 is shown.

20. Set the signal range of the analog input.

See table Analog inputs in chapter [6.5 Group 3.5: I/O Configuration](#).

21. Set a value for P3.13.2.1 (Keypad setpoint 1) and P3.13.2.2 (Keypad setpoint 2).
22. Using the sleep function.

➡ If the value "Yes" is given step 22, the next 3 steps are shown.

23. Set a value for P3.13.5.1 SP1 Sleep Frequency Limit.
24. Set a value for P3.13.5.2 SP1 Sleep Delay.
25. Set a value for P3.13.5.3 SP1 Wake-up Level.
26. Set a value for P3.15.2 Number of pumps.
27. Set a value for P3.15.5 Pump Interlocking.
28. Set a value for P3.15.6 Autochange.

➡ If the value Enabled (Interval or Weekdays) is set to parameter Autochange, the next 6 steps are shown. If the Autochange function is not used, the wizard goes directly to step 21.

29. Set a value for P3.15.7 Autochanged pumps.

➡ If the value Enabled (Interval) is set to parameter Autochange in step 28, the next step is shown.

30. Set a value for P3.15.8 Autochange Interval.

➡ If the value Enabled (Weekdays) is set to parameter Autochange in step 28, the next 2 steps are shown.

31. Set a value for P3.15.9 Autochange Days.
32. Set a value for P3.15.10 Autochange Time of Day.
33. Set a value for P3.15.11 Autochange Frequency Limit.
34. Set a value for P3.15.12 Autochange Pump Limit.
35. Set a value for P3.15.13 Bandwidth.
36. Set a value for P3.15.14 Bandwidth Delay.

➡ The Multi-pump (single drive) application wizard is completed.

4.1.5 Multi-pump (Multidrive) Application Wizard

The application wizard helps to set the basic parameters that are related to the application.

To start the Multi-pump (single drive) application wizard, set the value PID Control to parameter P1.2 Application (ID 212) in the keypad.

If the application wizard is started from the Start-up wizard, the wizard goes directly to step 11.

See the range of values of the parameters in [6.1 Group 3.1: Motor Settings](#), [6.3 Group 3.3: References](#), [6.4 Group 3.4: Ramps and Brakes Setup](#), [6.13 Group 3.13: PID Controller](#).

Procedure

1. Set a value for P3.1.2.2 Motor Type (so that it agrees with the nameplate of the motor).

➡ If Motor Type is set to Induction Motor, also step 6 is shown. If the selection is PM Motor, the value of parameter P3.1.1.5 Motor Cos Phi is set to 1.00, and the wizard skips step 6.

2. Set a value for P3.1.1.1 Motor Nominal Voltage (so that it agrees with the nameplate of the motor).
3. Set a value for P3.1.1.2 Motor Nominal Frequency (so that it agrees with the nameplate of the motor).
4. Set a value for P3.1.1.3 Motor Nominal Speed (so that it agrees with the nameplate of the motor).
5. Set a value for P3.1.1.4 Motor Nominal Current (so that it agrees with the nameplate of the motor).
6. Set a value for P3.1.1.5 Motor Cos Phi (so that it agrees with the nameplate of the motor).
7. Set a value for P3.3.1.1 Minimum Frequency Reference.
8. Set a value for P3.3.1.2 Maximum Frequency Reference.
9. Set a value for P3.4.1.2 Acceleration Time 1.
10. Set a value for P3.4.1.3 Deceleration Time 1.
11. Select a control place (where to give the start and stop commands, and the frequency reference of the drive).
12. Set a value for P3.13.1.4 Process Unit Selection.

➡ If the selection is other than %, the next steps are shown. If the selection is %, the wizard goes directly to step 16.

13. Set a value for P3.13.1.5 Process Unit Min.
14. Set a value for P3.13.1.6 Process Unit Max.
15. Set a value for P3.13.1.7 Process Unit Decimals.
16. Set a value for P3.13.3.3 Feedback 1 Source Selection.

See the table Feedback settings in chapter [6.13 Group 3.13: PID Controller](#).

➡ If analog input signal is selected, step 17 is shown. With other selections, the wizard goes to step 18.

17. Set the signal range of the analog input.

See table Analog inputs in chapter [6.5 Group 3.5: I/O Configuration](#).

18. Set a value for P3.13.1.8 Error Inversion.
19. Set a value for P3.13.2.6 Setpoint Source Selection.

See table Setpoints in chapter [6.13 Group 3.13: PID Controller](#).

➡ If analog input signal is selected, step 20 is shown. With other selections, the wizard goes to step 21. If Keypad Setpoint 1 or Keypad Setpoint 2 is selected for the value, step 22 is shown.

20. Set the signal range of the analog input.

See table Analog inputs in chapter [6.5 Group 3.5: I/O Configuration](#).

21. Set a value for P3.13.2.1 (Keypad setpoint 1) and P3.13.2.2 (Keypad setpoint 2).
22. Using the sleep function.

➡ If the value "Yes" is given step 22, the next 3 steps are shown.

23. Set a value for P3.13.5.1 SP1 Sleep Frequency Limit.
24. Set a value for P3.13.5.2 SP1 Sleep Delay.
25. Set a value for P3.13.5.3 SP1 Wake-up Level.
26. Set a value for P3.15.1 Multi-pump Mode.
27. Set a value for P3.15.3 Pump ID Number.
28. Set a value for P3.15.4 Start and Feedback.
29. Set a value for P3.15.2 Number of pumps.
30. Set a value for P3.15.5 Pump Interlocking.

31. Set a value for P3.15.6 Autochange.

➡ If the value Enabled (Interval) is set to parameter Autochange, step 33 shows. If the value Enabled (Weekdays) is set to parameter Autochange, step 34 shows. If the Autochange function is not used, the wizard goes directly to step 36.

32. Set a value for P3.15.7 Autochanged pumps.

➡ If the value Enabled (Interval) is set to parameter Autochange in step 31, the next step is shown.

33. Set a value for P3.15.8 Autochange Interval.

➡ If the value Enabled (Weekdays) is set to parameter Autochange in step 31, the next 2 steps are shown.

34. Set a value for P3.15.9 Autochange Days.

35. Set a value for P3.15.10 Autochange Time of Day.

36. Set a value for P3.15.13 Bandwidth.

37. Set a value for P3.15.14 Bandwidth Delay.

➡ The Multi-pump (single drive) application wizard is completed.

4.2 Function Wizards

4.2.1 Wizards in Menu 1.1

Table 12: M1.1 Wizards

Index	Parameter	Min	Max	Unit	Default	ID	Description
1.1.1	Start-up wizard	0	1		0	1171	0 = Do not activate 1 = Activate The selection Activate starts the Start-up wizard (see 1.6.1 Start-Up Wizard).
1.1.2	Fire mode Wizard	0	1		0	1672	The selection Activate starts the Fire mode wizard (see 4.2.2 Fire Mode Wizard).

4.2.2 Fire Mode Wizard

To start the Fire Mode Wizard, make the selection Activate for parameter B1.1.4 in the Quick setup menu. See [11.19.1 The Fire Mode Function](#) for information on the password and warranty.

See the range of values of the parameters in [6.17 Group 3.17: Fire Mode](#).

NOTICE

Before continuing, read about the password and warranty.

Procedure

1. Set a value for parameter P3.17.2 Fire Mode Frequency Source.

➡ If other value than Fire mode frequency is selected, the wizard goes directly to step 3.

2. Set a value for parameter P3.17.3 Fire Mode Frequency.

3. Activate the signal when the contact opens or when it closes.

4. Set a value for parameters P3.17.4 Fire Mode Activation on OPEN/P3.17.5 Fire Mode Activation on CLOSE.

See also chapter [11.7.1.1 General Information on the Programming](#).

5. Set a value for parameter P3.17.6 Fire Mode Reverse.

6. Set a value for P3.17.1 Fire Mode Password.

5 Monitor Menu

5.1 Uses of the Monitor Menu

Monitor Menu allows monitoring the actual values of the parameters and signals. It also allows monitoring the statuses and measurements. Some of the values to be monitored can be customized.

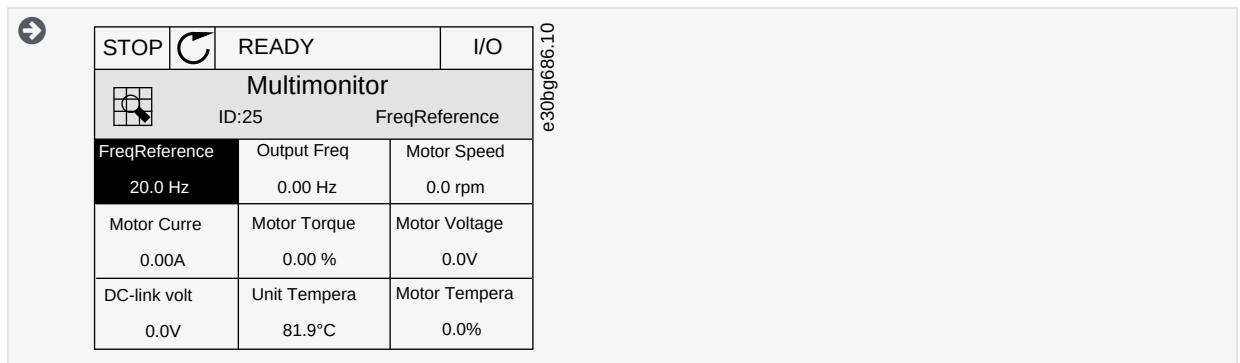
5.2 Multimonitor Monitoring

5.2.1 Changing the Items to Monitor

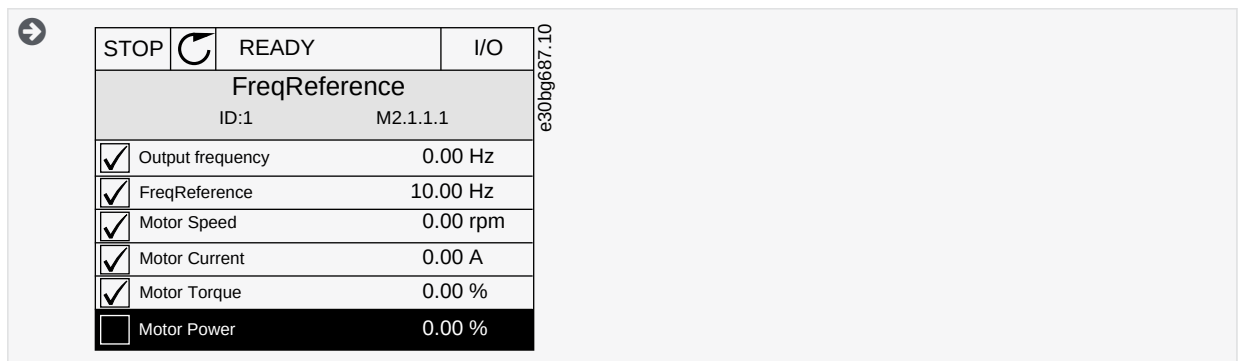
On the *Multimonitor* page, 4–9 items to monitor can be collected. Select the number of items with the parameter 3.11.4 Multimonitor View. See more in [6.11 Group 3.11: Application Settings](#).

Procedure

1. Go into the *Monitor* menu with the [OK] button.
2. Go into *Multimonitor*.
3. To replace an old item, activate it. Use the arrow buttons.



4. To select a new item in the list, push [OK].



5.3 Trend Curve Monitoring

5.3.1 Uses of the Trend Curve

The Trend curve is a graphical presentation of 2 monitor values.

When a value is selected, the drive starts to record the values. The *Trend curve* submenu allows examining the trend curve, and making the signal selections. Also the minimum and maximum settings and the sampling interval, and use autoscaling can be given there.

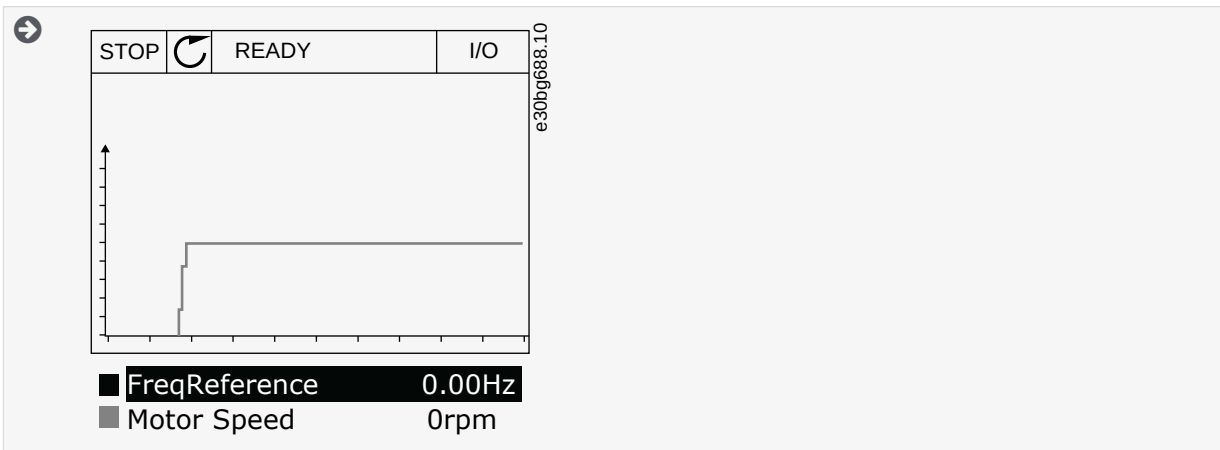
5.3.2 Changing the Values in Trend Curve Monitoring

Procedure

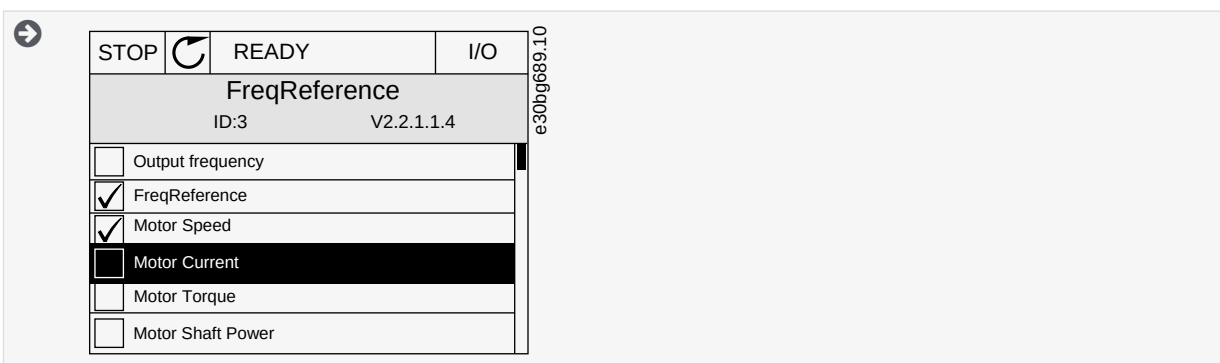
1. In the *Monitor* menu, find the *Trend curve* submenu and push [OK].
2. Go into the submenu *View trend curve* with the [OK] button.

- To select the current value that to change, use the arrow buttons up and down. Push [OK].

Only 2 values as trend curves can be monitored at the same time. The current selections, *FreqReference*, and *Motor speed*, are at the bottom of the display.



- Go through the list of the monitoring values with the arrow buttons.



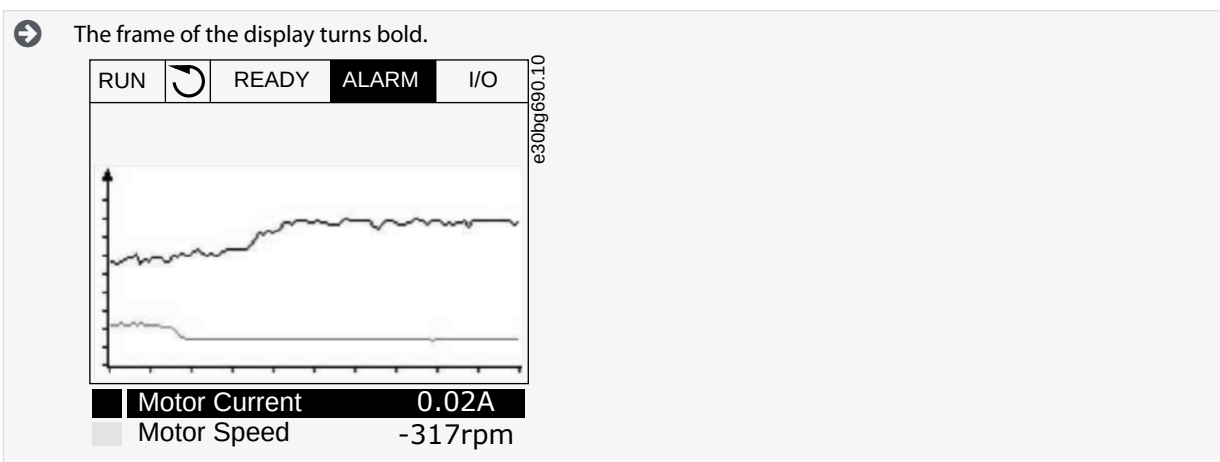
- Make a selection and push [OK].

5.3.3 Stopping the Progression of the Curve

The *Trend curve* function allows stopping the curve and reading the current values. After, the progression of the curve can be restarted.

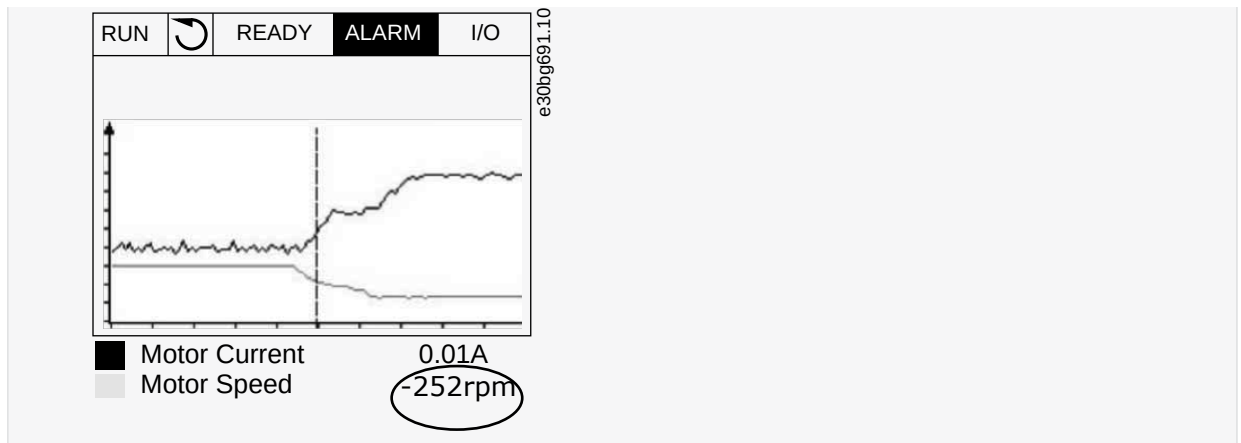
Procedure

- In *Trend curve* view, activate a curve with the arrow button Up.



- Push [OK] at the target point of the curve.

A vertical line comes into view on the display. The values at the bottom of the display agree to the location of the line.



3. To move the line to see the values of some other location, use the arrow buttons Left and Right.

5.3.4 Trend Curve Parameters

Table 13: The Trend Curve Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
M2.2.1	View Trend curve						Go into this menu to monitor values in a curve form.
P2.2.2	Sampling interval	100	432000	ms	100	2368	
P2.2.3	Channel 1 min	-214748	1000		-1000	2369	
P2.2.4	Channel 1 max	-1000	214748		1000	2370	
P2.2.5	Channel 2 min	-214748	1000		-1000	2371	
P2.2.6	Channel 2 max	-1000	214748		1000	2372	
P2.2.7	Autoscale	0	1		0	2373	0 = Disabled 1 = Enabled

5.4 Basic Monitoring

5.4.1 Uses of the Basic Monitoring

Only the standard I/O board statuses are available in the Monitor menu. For the statuses of all the I/O board signals as raw data, see the I/O and Hardware menu.

Check the statuses of the expander I/O board in the I/O and Hardware menu when the system asks.

5.4.2 Basic Monitoring Parameters

Table 14: Items in the Basic Monitoring Menu

Index	Monitoring value	Unit	Scale	ID	Description
V2.3.1	Output frequency	Hz	0.01	1	
V2.3.2	Frequency reference	Hz	0.01	25	
V2.3.3	Motor speed	RPM	1	2	
V2.3.4	Motor current	A	Varies	3	
V2.3.5	Motor torque	%	0.1	4	
V2.3.7	Motor shaft power	%	0.1	5	

Index	Monitoring value	Unit	Scale	ID	Description
V2.3.8	Motor shaft power	kW/hp	Varies	73	
V2.3.9	Motor voltage	V	0.1	6	
V2.3.10	DC-link voltage	V	1	7	
V2.3.11	Unit temperature	°C	0.1	8	
V2.3.12	Motor temperature	%	0.1	9	
V2.3.13	Motor Preheat		1	1228	0 = OFF 1 = Heating (feeding DC-current)
V2.3.15	kWh Trip Counter Low	kWh	1	1054	
V2.3.14	kWh Trip Counter High		1	1067	

5.5 I/O Monitoring

Table 15: I/O Signal Monitoring

Index	Monitoring value	Unit	Scale	ID	Description
V2.4.1	Slot A DIN 1, 2, 3		1	15	
V2.4.2	Slot A DIN 4, 5, 6		1	16	
V2.4.3	Slot B RO 1, 2, 3		1	17	
V2.4.4	Analog input 1	%	0.01	59	Slot A.1 as default.
V2.4.5	Analog input 2	%	0.01	60	Slot A.2 as default.
V2.4.6	Analog input 3	%	0.01	61	Slot D.1 as default.
V2.4.7	Analog input 4	%	0.01	62	Slot D.2 as default.
V2.4.8	Analog input 5	%	0.01	75	Slot E.1 as default.
V2.4.9	Analog input 6	%	0.01	76	Slot E.2 as default.
V2.4.10	Slot A AO1	%	0.01	81	

5.6 Temperature Inputs

Table 16: Monitoring the Temperature Inputs

Index	Monitoring value	Unit	Scale	ID	Description
V2.5.1 ⁽¹⁾	Temperature input 1	°C	0.1	50	
V2.5.2 ⁽¹⁾	Temperature input 2	°C	0.1	51	
V2.5.3 ⁽¹⁾	Temperature input 3	°C	0.1	52	

Index	Monitoring value	Unit	Scale	ID	Description
V2.5.4 ⁽¹⁾	Temperature input 4	°C	0.1	69	
V2.5.5 ⁽¹⁾	Temperature input 5	°C	0.1	70	
V2.5.6 ⁽¹⁾	Temperature input 6	°C	0.1	71	

¹ This parameter group is visible with an option board for temperature measurement (OPTBH).

5.7 Extras and Advanced

Table 17: Monitoring of the Advanced Values

Index	Monitoring value	Unit	Scale	ID	Description
V2.6.1	Drive Status Word		1	43	B1 = Ready B2 = Run B3 = Fault B6 = Run Enable B7 = Alarm Active B10 = DC current in stop B11 = DC brake active B12 = Run Request B13 = Motor Regulator Active B15 = Brake Chopper Active
V2.6.2	Ready status		1	78	B0 = Run Enable high B1 = No fault active B2 = Charge switch closed B3 = DC voltage OK B4 = Power unit OK B5 = Power unit is not blocking start B6 = System software is not blocking start
V2.6.3	Application Status Word1		1	89	B0 = Interlock 1 B1 = Interlock 2 B2 = Ramp 2 active B3 = Reserved B4 = I/O A control active B5 = I/O B control active B6 = Fieldbus Control Active B7 = Local control active B8 = PC control active B9 = Preset frequencies active B10 = Flushing active B11 = Fire Mode active B12 = Motor Preheat active B13 = Quick stop active B14 = Drive stopped from keypad
V2.6.4	Application Status Word2		1	90	B0 = Acc/Dec prohibited B1 = Motor switch open B2 = PID running B3 = PID sleep

Index	Monitoring value	Unit	Scale	ID	Description
					B4 = PID soft fil B5 = Autocleaning active B6 = Jockey pump B7 = Priming pump B8 = Anti-blocking B9 = Input pressure alarm B10 = Frost protection alarm B11 = Overpressure alarm B14 = Supervision 1 B15 = Supervision 2
V2.6.5	DIN Status Word 1		1	56	
V2.6.6	DIN Status Word 2		1	57	
V2.6.7	Motor current 1 decimal		0.1	45	
V2.6.8	Frequency reference source		1	1495	0 = PC 1 = Preset Freqs 2 = Keypad Reference 3 = Fieldbus 4 = AI1 5 = AI2 6 = AI1+AI2 7 = PID Controller 8 = Motor Potentiom. 10 = Flushing 11 = Block Out.1 12 = Block Out.2 13 = Block Out.3 14 = Block Out.4 15 = Block Out.5 16 = Block Out.6 17 = Block Out.7 18 = Block Out.8 19 = Block Out.9 20 = Block Out.10 100 = Not defined 101 = Alarm,PresetFreq 102 = Auto-cleaning
V2.6.9	Last active fault code		1	37	
V2.6.10	Last active fault ID		1	95	
V2.6.11	Last active alarm code		1	74	
V2.6.12	Last active alarm ID		1	94	
V2.6.13	Motor regulator status		1	77	B0 = Current Limit (Motor) B1 = Current Limit (Generator) B2 = Torque Limit (Motor) B3 = Torque Limit (Generator) B4 = Overvoltage Control

Index	Monitoring value	Unit	Scale	ID	Description
					B5 = Undervoltage Control B6 = Power Limit (Motor) B7 = Power Limit (Generator)
V2.6.14	Motor Shaft Power 1 Decimal	kW/hp		98	

5.8 Timer Functions Monitoring

Table 18: Monitoring of the Timer Functions

Index	Monitoring value	Unit	Scale	ID	Description
V2.7.1	TC 1, TC 2, TC 3		1	1441	
V2.7.2	Interval 1		1	1442	
V2.7.3	Interval 2		1	1443	
V2.7.4	Interval 3		1	1444	
V2.7.5	Interval 4		1	1445	
V2.7.6	Interval 5		1	1446	
V2.7.7	Timer 1	s	1	1447	
V2.7.8	Timer 2	s	1	1448	
V2.7.9	Timer 3	s	1	1449	
V2.7.10	Real-time clock			1450	

5.9 PID Controller Monitoring

Table 19: Monitoring of the Values of the PID Controller

Index	Monitoring value	Unit	Scale	ID	Description
V2.8.1	PID setpoint	Varies	As is set in P3.13.1.7 (See 6.13 Group 3.13: PID Controller.)	20	
V2.8.2	PID feedback	Varies	As is set in P3.13.1.7	21	
V2.8.3	PID Feedback (1)	Varies	As is set in P3.13.1.7	15541	
V2.8.4	PID Feedback (2)	Varies	As is set in P3.13.1.7	15542	
V2.8.5	PID Error	Varies	As is set in P3.13.1.7	22	
V2.8.6	PID Output	%	0.01	23	
V2.8.7	PID Status		1	24	0 = Stopped 1 = Running 3 = Sleep mode 4 = In dead band (see 6.13 Group 3.13: PID Controller)

5.10 External PID Controller Monitoring

Table 20: Monitoring of the Values of the External PID Controller

Index	Monitoring value	Unit	Scale	ID	Description
V2.9.1	ExtPID Setpoint	Varies	As set in P3.14.1.10 (See 6.14 Group 3.14: External PID Controller.)	83	
V2.9.2	ExtPID Feedback	Varies	As set in P3.14.1.10	84	
V2.9.3	ExtPID Error	Varies	As set in P3.14.1.10	85	
V2.9.4	ExtPID Output	%	0.01	86	
V2.9.5	ExtPID Status		1	87	0 = Stopped 1 = Running 2 = In dead band (See 6.14 Group 3.14: External PID Controller.)

5.11 Multi-pump Monitoring

Table 21: Multi-pump Monitoring

Index	Monitoring value	Unit	Scale	ID	Description
V2.10.1	Motors Running		1	30	
V2.10.2	Autochange		1	1114	
V2.10.3	Next Autochange	h	0.1	1503	
V2.10.4	Operate Mode		1	1505	0 = Slave 1 = Master
V2.10.5	Multi-pump Status		1	1628	0 = Not used 10 = Stopped 20 = Sleep 30 = Anti-blocking 40 = Auto-cleaning 50 = Flushing 60 = Soft filling 70 = Regulating 80 = Following 90 = Const. producing 200 = Unknown
V2.10.6	CommunicationStatus	h	0.1	1629	0 = Not used (Multi-pump multidrive function) 10 = Fatal communication errors occurred (or no communication) 11 = Errors occurred (data sending) 12 = Errors occurred (data receiving) 20 = Communication operational, no errors occurred 30 = Status unknown
V2.10.7	Pump (1) RunningTime	h	0.1	1620	
V2.10.8	Pump 2 RunningTime	h	0.1	1621	
V2.10.9	Pump 3 RunningTime	h	0.1	1622	
V2.10.10	Pump 4 RunningTime	h	0.1	1623	
V2.10.11	Pump 5 RunningTime	h	0.1	1624	